

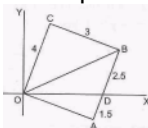
Miscellaneous Questions

Type – 1

Choose the most appropriate option (a, b, c or d).

- Q 1. If y is a function of x defined by $a^{x+y} = a^x + a^y$ where a is a real constant ($a > 1$) then the domain of $y(x)$ is
 (a) $(0, +\infty)$ (b) $(-\infty, 0)$ (c) $(-1, +\infty)$ (d) $(-\infty, 1)$
- Q 2. If y is a function of x given by $21\log y - \log x - \log(y-1) = 0$ then the
 (a) domain = $[4, +\infty)$, range = $(1, +\infty)$ (b) domain = $[4, +\infty)$, range = $(2, +\infty)$
 (c) domain = $(2, +\infty)$, range = $(2, +\infty)$ (d) none of these
- Q 3. If the latus rectum of the parabola $2x^2 - ky + 3 = 0$ be 2 then the vertex is
 (a) $\left(0, \frac{3}{4}\right)$ (b) $\left(0, \frac{3}{2}\right)$ (c) $\left(\frac{3}{2}, 0\right)$ (d) $(0, 0)$
- Q 4. The ends of a quadrant of a circle have the coordinates $(1, 3)$ and $(3, 1)$. Then the centre of such a circle is
 (a) $(2, 2)$ (b) $(1, 1)$ (c) $(4, 4)$ (d) $(2, 6)$
- Q 5. If the function $f(x) = x^2 - ax + 4$ is monotonic increasing in the open interval $(2, +\infty)$ then a is
 (a) 2 (b) 4 (c) 1 (d) $1/2$
- Q 6. If $\begin{vmatrix} x & 4 & 7 & 9 \\ 0 & 1 & 5 & 8 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \end{vmatrix}$ = the total number of arrangements of 4 different things taking at least one at a time, then x is equal to
 (a) $\frac{11}{8}$ (b) 4 (c) $\frac{11}{2}$ (d) $\frac{32}{3}$
- Q 7. Let A_r ; $r = 1, 2, 3, \dots$, be points on the number line such that $OA_1, OA_2, OA_3, \dots$ are in GP where O is the origin, and the common ratio of the GP be a positive proper fraction. Let M_r be the middle point of the line segment $A_{r+1}A_r$.
 Then the value of $\sum_{r=1}^{\infty} OM_r$ is equal to
 (a) $\frac{OA_1(OA_1 - OA_2)}{2(OA_1 + OA_2)}$ (b) $\frac{OA_1(OA_1 + OA_2)}{2(OA_1 - OA_2)}$ (c) $\frac{OA_1}{2(OA_1 - OA_2)}$ (d) ∞
- Q 8. If $x^2 - px + q$, $q \neq 0$, is a factor of $x^3 - ax^2 + bx - c$, $c \neq 0$, then
 (a) $pq + c = aq$ (b) $q(ac - b + 1) = c^2$ (c) $q^2 + c^2 = pqc$ (d) $a^2 + b^2 + c^2 = p^2 + q^2$
- Q 9. The shortest distance between the lines whose equations are $\vec{r} = t(\vec{i} + \vec{j} + \vec{k})$ and $\vec{r} = \vec{k} + s(\vec{i} - 2\vec{j} + 3\vec{k})$ is
 (a) 3 (b) $\frac{3}{\sqrt{38}}$ (c) $\sqrt{\frac{3}{14}}$ (d) $\frac{2}{\sqrt{13}}$
- Q 10. Four cards are drawn at random from a pack of 52 playing cards. The probability that the draw contains exactly one pair (i.e., having the same number) is
 (a) $\frac{13 \times {}^{12}C_2 \times 4^2}{{}^{52}C_4}$ (b) $\frac{13 \times {}^4C_2 \times {}^{12}C_2 \times 4^2}{{}^{52}C_4}$ (c) $\frac{13 \times {}^4C_2 \times {}^{48}C_2}{{}^{52}C_4}$ (d) none of these
- Q 11. If $\frac{1}{\sqrt{2}} < x < 1$ then $\cos^{-1}x + \cos^{-1}\left(\frac{x + \sqrt{1-x^2}}{\sqrt{2}}\right)$ is equal to
 (a) $2\cos^{-1}x - \frac{\pi}{4}$ (b) $2\cos^{-1}x$ (c) $\frac{\pi}{4}$ (d) 0
- Q 12. Two circles $x^2 + y^2 - 2kx = 0$ and $x^2 + y^2 - 4x - 8y + 16 = 0$ touch each other externally. Then k is

- Q 13. ABC is an equilateral triangle of side 4 cm. If R, r, h be the circumradius inradius and altitude respectively then $\frac{R+r}{h}$ is equal to
 (a) 4 (b) 1 (c) 2 (d) -4
- Q 14. If $y = 2$ be the directrix and $(0, 1)$ be the vertex of the parabola $x^2 + \lambda y + \mu = 0$ then (λ, μ) is equal to
 (a) 4 (b) 2 (c) 1 (d) 3
- Q 15. The real value of a for which the value of m satisfying the equation $(a^2 - 1)m^2 - (2a - 3)m + a = 0$ gives the slope of a line parallel to the y-axis
 (a) (8,-8) (b) (8,8) (c) (-4,4) (d) (4,-4)
- Q 16. The number of values of θ satisfying $4\cos \theta + 3 \sin \theta = 5$ as well as $3\cos \theta + 4\sin \theta = 5$ is
 (a) $\frac{3}{2}$ (b) 0 (c) 1 (d) ± 1
- Q 17. Four boys and eight girls sit at random in a row. The probability that each boy sits between two girls is
 (a) one (b) two (c) zero. (d) none of these
- Q 18. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 3[x]}{[x]}$, where $[x]$ denotes the greatest integer $\leq x$, is equal to
 (a) $\frac{7! \times 8!}{3! \times 12!}$ (b) $\frac{{}^7C_3}{12!}$ (c) $\frac{{}^7C_3 \times 8!}{12!}$ (d) none of these
- Q 19. If $\int_{a-n}^n e^{x(a-x)} dx = \lambda$ then value of $\int_{a-n}^n x e^{x(a-x)} dx, a \neq 2n$, is
 (a) $\frac{a\lambda}{2}$ (b) $a\lambda$ (c) $2a\lambda$ (d) none of these
- Q 20. The value of $\tan^{-1} 2 + \tan^{-1} 3$ is equal to
 (a) $-\frac{\pi}{4}$ (b) $\frac{3\pi}{4}$ (c) $\tan^{-1} 5$ (d) $\frac{\pi}{4}$
- Q 21. If in the $\triangle ABC$, AC is double of AB then the value of $\cot \frac{A}{2} \cdot \cot \frac{B-C}{2}$ is equal to
 (a) $\frac{1}{3}$ (b) $-\frac{1}{3}$ (c) 3 (d) $\frac{1}{2}$
- Q 22. Let $C_1, C_2, \dots, C_n, \dots$ be a sequence of concentric circles. The n th circle has the radius n and it has n openings. A point P starts travelling on the smallest circle C_1 and leaves it at an opening along the normal at the point of opening to reach the next circle C_2 . Then it moves on the second circle C_2 and leaves it likewise to reach the third circle C_3 and so on. The total number of different paths in which the point can come out of the n th circle is
 (a) $2^n \cdot n!$ (b) $2^{n-1} \cdot n!$ (c) $n!$ (d) $2^{n-1} \cdot (n-1)!$
- Q 23. If $\vec{a}, \vec{b}$ are orthogonal unit vectors then for a vector $\vec{r}$, noncoplanar with $\vec{a}$ and $\vec{b}$, the vector $\vec{r} \times \vec{a}$ is equal to
 (a) $[\vec{r} \vec{a} \vec{b}] \vec{b} + (\vec{r} \cdot \vec{b}) \vec{b} \times \vec{a}$ (b) $[\vec{r} \vec{a} \vec{b}] (\vec{a} + \vec{b})$ (c) $[\vec{r} \vec{a} \vec{b}] \vec{a} + (\vec{r} \cdot \vec{a}) \vec{a} \times \vec{b}$ (d) none of these
- Q 24. If the rank of the matrix $\begin{bmatrix} 4 & 2 & 1-x \\ 5 & k & 1 \\ 6 & 3 & 1+x \end{bmatrix}$ is 2 then
 (a) $k = \frac{5}{2}, x = \frac{1}{5}$ (b) $k = \frac{5}{2}, x \neq \frac{1}{5}$ (c) $k = \frac{1}{5}, x = \frac{5}{2}$ (d) none of these

- Q 25. The number of solutions of the equation $\int_{-2}^x |\cos x| dx = 0, 0 < x < \frac{\pi}{2}$, is
 (a) 0 (b) 1 (c) 2 (d) 4
- Q 26. If the line $ax + by = 2$ is a normal to the circle $x^2 + y^2 - 4x - 4y = 0$ and a tangent to the circle $x^2 + y^2 = 1$ then
- Q 27. If A and B are two independent events then which of the following is not equal to any of the remaining?
 (a) $P(A' \cap B') - P(A \cap B)$ (b) $P(A') + P(B') - 1$ (c) $P(B) - P(A')$ (d) $P(B') - P(A)$
- Q 28. An event X can take place in conjunction with any one of the mutually exclusive and exhaustive events A, B and C. If A, B, C are equiprobable and the probability of X is $\frac{5}{12}$, and the probability of X taking place when A has happened is $\frac{3}{8}$ while it is $\frac{1}{4}$ when B has taken place then the probability of X taking place in conjunction with C is
 (a) $\frac{5}{8}$ (b) $\frac{3}{8}$ (c) $\frac{5}{24}$ (d) none of these
- Q 29. The largest value of a for which the circle $x^2 + y^2 = a^2$ falls totally in the interior of the parabola $y^2 = 4(x + 4)$ is
 (a) $4\sqrt{3}$ (b) 4 (c) $4\sqrt{\frac{6}{7}}$ (d) $2\sqrt{3}$
- Q 30. Let $\vec{f}(t) = [t] \vec{i} + \{t - [t]\} \vec{j} + [t + 1] \vec{k}$, where $[.]$ denotes the greatest integer function. Then the vectors $\vec{f}\left(\frac{5}{4}\right)$ and $\vec{f}(t), 0 < t < 1$, are
 (a) perpendicular to each other (b) parallel to each other
 (c) inclined at an angle $\cos^{-1} \frac{2}{\sqrt{5(1+t^2)}}$ (d) inclined at $\cos^{-1} \frac{8+t}{9\sqrt{1+t^2}}$
- Q 31. The equation of the diagonal OB of the rectangle given in the diagram is

 (a) $y = 2x$ (b) $2y = x$ (c) $3y = x$ (d) none of these
- Q 32. Let $f(x)$ be a continuous function whose range is $[2, 6.5]$. If $h(x) = \left[\frac{\cos x + f(x)}{\lambda} \right]$, $\lambda \in \mathbb{N}$, be continuous, where $[.]$ denotes the greatest integer function, then the least value of X is
 (a) 6 (b) 7 (c) 8 (d) none of these
- Q 33. For the equation $1 + \sin x = ax$, the number of solutions is
 (a) 1 if $a > 1$ (b) at least 3 if $0 < a < 1$ (c) 2 if $a = 1$ (d) none of these
- Q 34. If the point $(0, \lambda)$ is a point of the region bounded by the semicircle $x = \sqrt{9 - y^2}$ and the line $x = 0$ then the equation $x^2 - 2x - \lambda = 0$ has
 (a) at least one real root in $[2, 3]$ (b) at least one real root in $[-1, 0]$
 (c) two real roots in $[-1, 3]$ (d) no real roots in $(0, 2)$
- Q 35. Let E be an event which is neither a certainty nor an impossibility. Its probability is such that $P(E) = 1 + \lambda + \lambda^2$ and $P(E') = (1 + \lambda)^2$ in terms of an unknown X. Then $P(E)$ is equal to
 (a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) none of these
- Q 36. A word of at least 5 letters is made at random from 3 vowels and 3 consonants, all the letters being different. The probability that no consonant falls between any two vowels in the word is
 (a) $\frac{9}{20}$ (b) $\frac{9}{10}$ (c) $\frac{7}{10}$ (d) $\frac{11}{10}$
- Q 37. A number is selected at random from the first twenty-five natural numbers. If it is a composite number then it is divided by 5. But if it is not a composite number, it is divided by 2. The probability that there will be no remainder in the division is
 (a) AP (b) 0.4 (c) 0.2 (d) none of these

- Q 38. If $P_r = (x_r, y_r)$; $r = 1, 2, 3$; are three points on the parabola $x^2 = 8y$, where x_1, x_2, x_3 are in GP, then y_1, y_2, y_3 are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 39. Let $f(x)$ be an even function in R . If $f(x)$ is monotonic increasing in $[2, 6]$ then
 (a) $f(3) > f(-5)$ (b) $f(-2) < f(2)$ (c) $f(-2) > f(2)$ (d) $f(-3) < f(5)$
- Q 40. If $(2) = g(2)$ and $f'(2) = a$, $g'(2) = 2a$ then $\frac{f(x) - g(x)}{x - 2}$ is equal to
 (a) 1 (b) a (c) $-a$ (d) none of these
- Q 41. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = |\vec{c}| = 1, |\vec{b}| = 4, |\vec{b} \times \vec{c}| = 2$ and $2\vec{b} = \vec{c} + \lambda \vec{a}$ then λ is
 (a) $\sqrt{65 - 8\sqrt{3}}$ (b) $\sqrt{17}$ (c) $\sqrt{3}$ (d) $\sqrt{\frac{17}{3}(2 + \sqrt{3})}$
- Q 42. If $f(x) = \int_0^x \sin[2x] dx$ where $[x]$ = greatest integer less than or equal to x then $f(\pi/2)$ is
 (a) $\frac{1}{2} \{\sin 1 + (\pi - 2) \sin 2\}$ (b) $\frac{1}{2} \{\sin 1 + \sin 2 + (\pi - 3) \sin 3\}$
 (c) 0 (d) $\sin 1 + \left(\frac{\pi}{2} - 2\right) \sin 2$
- Q 43. $\int_0^2 e^{x+[x]} d(x - [x])$, where $[x]$ = greatest integer $\leq x$, is equal to
 (a) $(e^2 - 1)(e + 1)$ (b) $e - 1$ (c) $(e^2 + 1)(e - 1)$ (d) $e^2 + 1$
- Q 44. Let $f(x) = \min\{x, 1 - x\}$ for all $x \in R$. Then the value of $\int f(x) dx$ is
 (a) $-\frac{1}{4}$ (b) $\frac{1}{4}$ (c) 2 (d) 0
- Q 45. A dice is thrown six times, being known that each time a different digit shows up. The probability a sum of 12 will be obtained in the first three throws is
 (a) $\frac{5}{24}$ (b) $\frac{25}{216}$ (c) $\frac{3}{20}$ (d) $\frac{1}{12}$
- Q 46. A composite number is selected at random from the first 30 natural numbers and it is divided by 5. The probability that there will be a remainder is
 (a) $\frac{14}{19}$ (b) $\frac{5}{19}$ (c) $\frac{5}{6}$ (d) $\frac{7}{15}$
- Q 47. In a $\triangle ABC$ the sides b, c and the angle B are given such that a has two? values a_1, a_2 . Then $|a_1 - a_2|$ is equal to
 (a) $\sqrt{b^2 - c^2 \sin^2 B}$ (b) $2\sqrt{b^2 - c^2 \sin^2 B}$ (c) $\sqrt{c^2 - b^2 \sin^2 B}$ (d) $2\sqrt{c^2 - b^2 \sin^2 B}$
- Q 48. If $f: (3, 4) \rightarrow (0, 1)$ is defined by $f(x) = x - [x]$ where $[x]$ denote the greatest integer function then $f^{-1}(x)$ is
 (a) $\frac{1}{x - [x]}$ (b) $[x] - x$ (c) $x - 3$ (d) $x + 3$
- Q 49. Let $f(x) = [4 + 3\cos x]$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, where $[x]$ = greatest integer less than or equal to x . The number of points of discontinuity of $f(x)$ is
 (a) 2 (b) 3 (c) 5 (d) none of these
- Q 50. Let $(x) = 1 - (1 - x)^2$, $x < 1$. The number of solutions of the equation $f\{f(x)\} = x$ is
 (a) one (b) two (c) zero (d) none of these
- Q 51. The system of inequations $x - y \leq -1$, $y \leq x$, $x \geq 0$, $y \geq 0$ has
 (a) one solution (b) infinite number of solutions
 (c) finite number of solutions (d) no solution
- Q 52. If $p^x = q^y = a$ and $q^y = z^z = b$ then
 (a) $xy = wz$ (b) $xz = yw$ (c) $x + z = y + w$ (d) $x + y = a + w$
- Q 53. If $\sin \theta - 1 = 0$ and $2\cos \theta + \sqrt{3} = 0$ then the general values of θ are given by

- (a) $2n\pi + \frac{\pi}{6}$ (b) $2n\pi + \frac{\pi}{4}$ (c) $2n\pi + \frac{7\pi}{6}$ (d) $2n\pi + \frac{5\pi}{6}$
- Q 54. The angle between the straight lines joining the point $(-1, 0)$ to the common points of $3x^2 + 5xy - 3y^2 + 8x + 8y + 5 = 0$ and $3x - 2y + 2 = 0$ is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) none of these
- Q 55. If $x \cos \alpha + y \sin \alpha = p$ is a tangent to the circle $x^2 + y^2 = 2q$ ($x \cos \alpha + y \sin \alpha$) then the set of possible values of p is
 (a) $\{q\}$ (b) $\{0, q\}$ (c) $\{0, 2q\}$ (d) $\{q, 2q\}$
- Q 56. If f and g are two functions such that $(fg)(x) = (gf)(x)$ for all x . Then f and g may be defined as
 (a) $f(x) = \sqrt{x}$, $g(x) = \cos x$
 (b) $f(x) = x^3$, $g(x) = x + 1$
 (c) $f(x) = x - 1$, $g(x) = x^2 + 1$
 (d) $f(x) = x^m$, $g(x) = x^n$ where m, n are unequal integers
- Q 57. If the parabola $y = ax^2 - 6x + b$ passes through $(0, 2)$ and has its tangent at $x = \frac{3}{2}$ parallel to the x -axis then
 (a) $a = 2$, $b = -2$ (b) $a = 2m$, $b = 2$ (c) $a = -2$, $b = 2$ (d) $a = -2$, $b = -$
- Q 58. If $f(x) = \sqrt{1 + \sqrt{x}}$, $x > 0$, then $f(x) \cdot f'(x)$ is equal to
 (a) $\frac{1}{2\sqrt{x}}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4\sqrt{x}}$ (d) $\frac{2\sqrt{x} + 1}{4\sqrt{x}}$
- Q 59. If in a $\triangle ABC$, $4 \tan A + 3 = 0$ then $\sin \frac{A}{2}$ and $\cos \frac{A}{2}$ are the roots of
 (a) $10x^2 - 4\sqrt{10}x + 3 = 0$ (b) $10x^2 + 4\sqrt{10}x + 3 = 0$
 (c) $101x^2 - 4\sqrt{10}x - 3 = 0$ (d) $10x^2 + 4\sqrt{10}x - 3 = 0$
- Q 60. If $\sin^6 \theta + \cos^6 \theta + k \cos^2 2\theta = 1$ then k is equal to
 (a) $\frac{1}{2} \tan^2 \theta$ (b) $\frac{1}{4} \tan^2 \theta$ (c) $4 \cot^2 \theta$ (d) $\frac{3}{4} \tan^2 2\theta$
- Q 61. If a, b, c are in AP and one root of the equation $ax^2 + bx + c = 0$ is 2 then the other roots is
 (a) $\frac{3}{4}$ (b) $-\frac{3}{4}$ (c) $-\frac{5}{4}$ (d) $-\frac{5}{2}$
- Q 62. The number of solution of $[\cos x] + |\sin x| = 1$ in $\pi \leq x < 3\pi$ is
 (a) 3 (b) 4 (c) 2 (d) 1
- Q 63. A variable line $\frac{x}{a} + \frac{y}{b} = 1$ is drawn through the point $(k, 2k)$ so as to form a triangle of area A . If a, b are of the same sign then the least value of A is
 (a) k^2 (b) $2k^2$ (c) $4k^2$ (d) $8k^2$
- Q 64. If the set $A = \left\{ x \mid \cos \frac{\pi}{x} > 0 \right\}$ then which of the following intervals is a subset of A ?
 (a) $\left(\frac{2}{5}, \frac{2}{3} \right)$ (b) $\left(\frac{2}{7}, \frac{2}{5} \right)$ (c) $\left(\frac{2}{7}, \frac{2}{3} \right)$ (d) $\left(\frac{2}{9}, \frac{2}{3} \right)$
- Q 65. If $\theta \in \left(0, \frac{\pi}{2} \right)$ then the value of $\cos \left(\theta - \frac{\pi}{4} \right)$ lies in the interval
 (a) $\left(\frac{1}{2}, 1 \right)$ (b) $\left(\frac{1}{\sqrt{2}}, 1 \right]$ (c) $\left(-\frac{1}{\sqrt{2}}, 1 \right]$ (d) $(0, 1)$
- Q 66. In $\triangle ABC$, the median AD divides $\angle BAC$ such that $\angle BAD : \angle CAD = 2 : 1$. Then $\cos \frac{A}{3}$ is equal to

- (a) $\frac{\sin B}{2 \sin C}$ (b) $\frac{\sin C}{2 \sin B}$ (c) $\frac{2 \sin B}{\sin C}$ (d) none of these
- Q 67. If the coordinates of two vertices and the orthocentre of a triangle are integers then the coordinates of the third vertex must be
(a) integers (b) rational numbers (c) irrational numbers (d) none of these
- Q 68. The number of triangles that the four lines $y = x + 3, y = 2x + 3, y = 3x + 2$ and $y + x = 3$ form is
(a) 4 (b) 2 (c) 3 (d) 1
- Q 69. Let PQR be a right-angled triangle, right angled at P(2,1). If the equation of the line QR is $2x + y = 3$ and Q = (2,-1) then the centroid of the ΔPQR
(a) (b) $\left(2, \frac{1}{3}\right)$ (c) (2, 0) (d) $\left(\frac{1}{3}, \frac{5}{3}\right)$
- Q 70. If n is a multiple of 3 then the coefficient of x^n in the expansion of $\log_e(1 + x + x^2)$, $|x| < 1$, is
(a) $-\frac{1}{n}$ (b) $\frac{2}{n}$ (c) $-\frac{2}{n}$ (d) $\frac{1}{n}$
- Q 71. The possible number of different orders that a matrix can have when it has 24 elements, is
(a) 8 (b) 16 (c) 4 (d) none of these
- Q 72. If in the expansion of $(1 + x)^m \cdot (1 - x)^n$, the coefficients of x and x^2 are 3 and -6 respectively then
(a) $m = 6, n = 15$ (b) $m = 9, n = 12$ (c) $m = 12, n = 9$ (d) $m = 24, n = 21$
- Q 73. The coefficient of x^{28} in the expansion of $(1 + x^3 - x^6)^{30}$ is
(a) 1 (b) 0 (c) ${}^{30}C_6$ (d) ${}^{30}C_3$
- Q 74. $(n + 2) \cdot {}^nC_0 \cdot 2^{n+1} - (n + 1) \cdot {}^nC_1 \cdot 2^n + n \cdot {}^nC_2 \cdot 2^{n-1} - \dots$ is equal to
(a) 4 (b) $4n$ (c) $4(n + 1)$ (d) $2(n + 2)$
- Q 75. If $u_n = 2 \cos n\theta$ then $u_1 u_n - u_{n-1}$ is equal to
(a) u_{n-2} (b) u_{n+1} (c) 0 (d) none of these
- Q 76. The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is
(a) 4 (b) 3 (c) 2 (d) 1
- Q 77. $\tan \frac{10\pi}{3}, \tan \frac{9\pi}{4}, \tan \frac{7\pi}{6}$ are in
(a) AP (b) GP (c) HP (d) none of these
- Q 78. The number of real solutions of the equation $\cos^7 x + \sin^5 x = 1$, $0 < x < 2\pi$, is
(a) 1 (b) 2 (c) 3 (d) infinite
- Q 79. The sum of $\cos^{-1} 3 + \cot^{-1} 13 + \cot^{-1} 21 + \cot^{-1} 31$ is
(a) $\cot^{-1} \frac{7}{5}$ (b) $\cot^{-1} \frac{5}{7}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$
- Q 80. One root of the equation $2 \cos x + 1 = 2x$ lies in the interval
(a) $\left[-\frac{\pi}{2}, 0\right]$ (b) $\left[0, \frac{\pi}{2}\right]$ (c) $\left[\frac{\pi}{2}, \pi\right]$ (d) none of these
- Q 81. If $2, 3^x - 4, \log_{3^x + \frac{7}{2}} 4$ are in HP then x is equal to
(a) 1 (b) 2 (c) 4 (d) 0
- Q 82. The value of $\sum_{r=1}^{\infty} \tan^{-1} \frac{2r}{2 + r^2 + r^4}$ is equal to
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) π
- Q 83. In a ΔPQR , $\angle R = \frac{\pi}{2}$. If $\tan \frac{P}{2}$ and $\tan \frac{Q}{2}$ are the roots of the equation $ax^2 + bx + c = 0$ then
(a) $a + b = c$ (b) $b + c = a$ (c) $a + c = b$ (d) $b = c$
- Q 84. A particle moving on a curve has the position given by $x = f(t) \sin t + f'(t) \cos t$, $y = f(t) \cos t - f'(t) \sin t$ where f is a thrice - differentiable function. Then the velocity of the particle at time t is
(a) $f'''(t)$ (b) $f'(t) + f'''(t)$ (c) $f'(t) + f''(t)$ (d) $f'(t) - f'''(t)$
85. If $f(x) = \cos^{-1} \frac{x - x^{-1}}{x + x^{-1}}$ then $f'(-2)$ is

- (a) $\frac{2}{5}$ (b) $\frac{-2}{5}$ (c) $-\frac{1}{5}$ (d) none of these
- Q 86. The slope of the tangent to the curve $\tan y = \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}$ at $x = \frac{1}{2}$ is
 (a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) 1 (d) $\frac{1}{2}$
- Q 87. $\frac{3x^3 + 5x - 4}{(x^2 - 1)(x^2 - 4)} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{x+2} + \frac{D}{x-2}$ then $A + B + C + D$ is
 (a) 4 (b) 3 (c) 0 (d) 2
- Q 88. The distance between the plane whose equation is $\vec{r} \cdot (2\vec{i} + \vec{j} + 3\vec{k}) = 5$ and the line whose equation is $\vec{r} = \vec{i} + \lambda(2\vec{i} + 5\vec{j} + 3\vec{k})$, is
 (a) $\frac{3}{\sqrt{14}}$ (b) $\frac{5}{\sqrt{14}}$ (c) 5 (d) 0
- Q 89. The number of ways in which 10 red balls, 10 green balls and a white ball can be arranged in a row with the white ball in the middle, and the arrangement of colours on balls being symmetrical about the white ball,
 (a) $\frac{2(10!)}{(5!)^2}$ (b) $\frac{10!}{(5!)^2}$ (c) $\frac{20!}{(10!)^2}$ (d) none of these
- Q 90. A flag is to be coloured in four stripes by using 6 different colours, no two consecutive stripes being of the same colour. This can be done in
 (a) 1500 ways (b) 750 ways (c) 6^4 ways (d) none of these
- Q 91. Two vertices of an equilateral triangle in the Argand plane represent the nonreal cube roots of unity. The area of the triangle is
 (a) $\frac{3\sqrt{3}}{4}$ (b) $\frac{3\sqrt{3}}{2}$ (c) $3\sqrt{3}$ (d) none of these
- Q 92. If $2x - y + 1 = 0$, $x + yz = 2$, $x > 0$, $y > 0$ then
 (a) $z \in [0, 2]$ (b) $z \in \left(-\frac{1}{2}, 0\right]$ (c) $z \in [0, 1]$ (d) $z \in \left(-\frac{1}{2}, 2\right)$
- Q 93. Let $A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and $a_{pq} = i^{p+q}$ where $i = \sqrt{-1}$. The value of A is
 (a) real and positive (b) real and negative (c) 0 (d) imaginary
- Q 94. A piece of paper is in the shape of a square of side 1 metre. It is cut at the four corners to make a regular polygon of eight sides (octagon). The area of the polygon is
 (a) $2(\sqrt{2} - 1)m^2$ (b) $(\sqrt{2} - 1)m^2$ (c) $\frac{1}{\sqrt{2}}m^2$ (d) none of these
- Q 95. The sum of the first n terms of the series $3 + \frac{5}{2} + \frac{9}{4} + \frac{17}{8} + \dots$ is
 (a) $2n + 2^{1-n}$ (b) $2\{N + 1 - 2^{-n}\}$ (c) $n + 2^n$ (d) none of these
- Q 96. The number of solutions of $||x - 4| - 1| = 2$ is
 (a) four (b) two (c) three (d) one
- Q 97. If $1 \in (\alpha, \beta)$ where α, β are roots of $x^2 - a(x + 1) + 3 = 0$ then
 (a) $a > 2$ (b) $a < -6$ or $a > 2$ (c) $-6 < a < 2$ (d) $a < 2$
- Q 98. The number of ways of selecting two numbers from the first 15 natural numbers such that their sum is a multiple of 5, is
 (a) 20 (b) 36 (c) 21 (d) 16
- Q 99. If $\left|z - \frac{2}{z}\right| = 1$ then the greatest value of $|z|$ is

- (a) 2 (b) 1 (c) 4 (d) 3
- Q 100. If $|z_1 - z_0| = |z_2 - z_0| = a$ and $\arg \frac{z_2 - z_0}{z_0 - z_1} = \frac{\pi}{2}$ then z_0 is equal to
 (a) $\frac{1}{2} \{(1+i)z_1 + (1-i)z_2\}$ (b) $\frac{1}{2} |(1-i)z_1 + (1+i)z_2|$ (c) $\frac{1}{2} |z_1 + z_2|$ (d) none of these
- Q 101. The fractional part of the number $\frac{3^{33}}{25}$ is
 (a) $\frac{3}{25}$ (b) $\frac{23}{25}$ (c) $\frac{22}{25}$ (d) $\frac{8}{25}$
- Q 102. A boy goes to a place with a speed of a km/h and comes back with a speed of b km/h. The constant speed with which he should travel to cover the entire distance in the same total time is
 (a) $\frac{a+b}{2}$ km/h (b) $\sqrt{ab}$ km/h (c) $\frac{2ab}{a+b}$ km/h (d) none of these
- Q 103. If a, B be unequal real roots of the equation $ax^2 + bx + c = 0$ where a, b, c are real and y is the solution of $2ax + b = 0$ then
 (a) $\gamma > \alpha, > \beta$ (b) $\gamma < \alpha, \gamma < \beta$ (c) $\alpha < \gamma < \beta$ or $\beta < \gamma < \alpha$ (d) none of these
- Q 104. If f be a function defined as $f(x) = x^3 - 3x, -1 \leq x \leq 3$ then the range of f is
 (a) $[0, 24]$ (b) $[-2, 2]$ (c) $[-2, 18]$ (d) none of these
- Q 105. Let a, b, c be unit vectors and α, β, γ are angles between the vectors a, b; b, c and c, a respectively. If $a + b + c$ is also a unit vector then $\cos a + \cos B + \cos y$ is equal to
 (a) -1 (b) 3 (c) -3 (d) 1
- Q 106. If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$ then the value of $xy + yz + zx$ is
 (a) -3 (b) 1 (c) 3 (d) none of these
- Q 107. Let $f_n(x) = \frac{n+1}{(n-1)!} x^n$. The value of $\int_0^1 \left(\sum_{n=1}^{\infty} f_n(x) \right) dx$ is
 (a) e (b) 0 (c) 2e (d) none of these
- Q 108. Let three positive numbers a, b, c be in GP such that a, b + 8, c are in AP, while a, b + 8, c + 64 are in GP. Then the AM of a, b, c is
 (a) $\frac{52}{3}$ (b) $\frac{28}{9}$ (c) 26 (d) 14
- Q 109. The number of ways in which the five digits 1,2,3,4,5 can be arranged to make a number greater than 10000 such that the odd digits are in the ascending order, is
 (a) 10 (b) 20 (c) 120 (d) 60
- Q 110. The number of real solutions of the equation $\left(\frac{3}{5}\right)^x + \frac{7}{5} = 2^x$ is
 (a) 2 (b) 1 (c) 0 (d) infinite
- Q 111. If $f(x) = x^3 - 6x^2 + 6x$ then $f(3|2 + \sqrt{4} + 2)$ has the value
 (a) 2 (b) 4 (c) 6 (d) 8
- Q 112. Let $f(x) = |x + 1|$. The number of values of $x \in [-2, 2]$ for which $f(x - 3), f(x - 1), f(x + 1)$ are in AT is
 (a) 1 (b) 2 (c) 0 (d) infinite
- Q 113. In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is
 (a) at least 30 (b) at most 20 (c) exactly 25 (d) none of these
- Q 114. If $(x + 1)(x^{1/2} + 1)(x^{1/2^2} + 1) = (x^2 - 1)f(x)$ then $f(x)$ is
 (a) $\frac{1}{\frac{1}{x^{2^{n-1}}} - 1}$ (b) $\frac{1}{\frac{1}{x^{2^n}} - 1}$ (c) $\frac{1}{\frac{1}{x^{2^n}} - 1}$ (d) none of these
- Q 115. The minimum positive integral value of x such that $(1073)^{71} - x$ is divisible by 10, is
 (a) 1 (b) 3 (c) 7 (d) 9
- Q 116. The minimum value of $|z - 1| + |z|$ for complex values of z is
 (a) 2 (b) $\frac{1}{2}$ (c) 0 (d) 1
- Q 117. If in $\triangle ABC, a^2 + b^2 > c^2$ then the A is

- (a) acute angled at C (b) obtuse angled at C (c) an acute-angled triangle (d) none of these
- Q 118. In the $\triangle ABC$, BC is produced to D and $\angle ACD = \frac{3\pi}{4}$ and $\tan A, \tan B$ are roots of the equation $x^2 - x + m = 0$. Then
 (a) $\lambda^2 - \mu^2 = 1 + 6\mu$ (b) $\lambda^2 - \mu^2 = 1$ (c) $\lambda = \mu - 1$ (d) none of these
- Q 119. If α, β are the roots of $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$ then $\frac{\alpha - \gamma}{\beta - \gamma} \cdot \frac{\alpha - \delta}{\beta - \delta}$ is equal to
 (a) 1 (b) -1 (c) $\frac{q+r}{p+r}$ (d) $\frac{q-r}{p-r}$
- Q 120. If a body starts with a velocity u in a straight line with uniform acceleration f and covers a distance s in time t seconds, and s_t denotes the distance covered by it in the t th second, then s_2, s_4, s_6 are in
 (a) AP (b) GP (c) the ratio 3 : 7 : 15 (d) the ratio 1 : 3 : 7
- Q 121. The value of $\sum_{r=1}^n r({}^nC_r + {}^rP_r)$ is
 (a) $n \cdot 2^{n-1} - 1$ (b) $n \cdot 2^{n-1} + (n+1)!$ (c) $n \cdot 2^{n-1} + (n+1)! - 1$ (d) $n^2 + n + 5$
- Q 222. For a positive integer n , let $a(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2^n - 1}$. Then
 (a) $a(100) \leq 100$ (b) $a(100) \geq 100$ (c) $a(200) \leq 100$ (d) $a(200) \geq 100$
- Q 123. If α is a nonreal root of $x^5 + 1 = 0$ then $\alpha^{110m+2} + \alpha^{5n+2} + \alpha^{5n}$, where n is an odd positive integer, has the value
 (a) 1 (b) 0 (c) -1 (d) none of these
- Q 124. If the 6th, 11th and 16th terms of a GP are a, b, c respectively then
 (a) $a + c = 2b$ (b) $b^2 = ac$ (c) $a^2 + c^2 = b^2$ (d) none of these
- Q 125. Let $f(x) = x^3 - 3x + 1$ and $f(0), f(1)$ are of opposite signs. Then the set of values of t is
 (a) $(0, 2)$ (b) $(-\infty, 0)$ (c) $(2, +\infty)$ (d) none of these
- Q 126. If $z = 1 + \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ then
 (a) $\operatorname{Re}(z^5) = \frac{\sqrt{3}}{2}$ (b) $\operatorname{Re}(z^5) = \frac{1}{2}$ (c) $\operatorname{Im}(z^5) = \frac{1}{2}$ (d) $\operatorname{Im}(z^5) = \frac{\sqrt{3}}{2}$
- Q 127. If $f(x) = \sum_{\lambda=1}^n \left(x - \frac{1}{\lambda}\right) \left(x - \frac{1}{\lambda+1}\right)$ then $\lim_{n \rightarrow \infty} f(0)$ is
 (a) 0 (b) -1 (c) 1 (d) none of these
- Q 128. If a, b, c, d are non-negative real numbers where $a + b + c + d = 1$ then the maximum value of $ab + bc + cd$ is
 (a) $\frac{1}{4}$ (b) 3 (c) 4 (d) none of these
- Q 129. If $x^3 - mx^2 - 3x + 2 = 0$ has two roots equal in magnitude but opposite in sign then m is
 (a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $-\frac{2}{3}$ (d) none of these
- Q 130. If $u_i = 1 - \frac{1}{i}$ then $u_2 \cdot u_3 \cdot \dots \cdot u_n$ is equal to
 (a) $\frac{1}{n}$ (b) $\frac{1}{n!}$ (c) 1 (d) none of these
- Q 131. The value of $\sum_{j=1}^n ({}^{n+1}C_j - {}^nC_j)$ is equal to
 (a) 2^n (b) $2^n + 1$ (c) $3 \cdot 2^n$ (d) $2^n - 1$
- Q 132. If in the triangle ABC the equation of the side BC is $x + 2y = 3$, the vertex A is $(1, 2)$ and the abscissa of the orthocentre is 3 then the ordinate of the orthocentre is
 (a) 6 (b) 4 (c) 2 (d) 0
- Q 133. If the circles $x^2 + y^2 - 2x = 0$ and $x^2 + y^2 + 2x + 4y = 4$ have only one common tangent then X is

- (a) 1 (b) -1 (c) 0 (d) 2
- Q 134. The least number of sides of a polygon in which the number of diagonals is at least 10 more than the number of sides, is
(a) 8 (b) 10 (c) 11 (d) 25
- Q 135. If the algebraic sum of the distance of the points (1,2), (-3,1) and (2, -5) from a straight line be equal to zero then the straight line must pass through
(a) (0,-2) (b) (0,0) (c) $\left(0, -\frac{2}{3}\right)$ (d) at least one of the given points
- Q 136. Let ABCDEF be a regular hexagon in the x-y plane and $\vec{AB} = 4\vec{AB} = 4\vec{i}$. Then CD is equal to
(a) $6\vec{i} + 2\sqrt{3}\vec{j}$ (b) $2(-\vec{i} + \sqrt{3}\vec{j})$ (c) $2(\vec{i} + \sqrt{3}\vec{j})$ (d) $2(\vec{i} - \sqrt{3}\vec{j})$
- Q 137. In a $\triangle ABC$, $\angle B = \frac{\pi}{3}$. The range of values of x , where $x = \sin A \cdot \sin C$, is the interval
(a) $\left[-\frac{1}{4}, \frac{3}{4}\right]$ (b) $\left(0, \frac{3}{4}\right)$ (c) $\left(0, \frac{3}{4}\right]$ (d) $\left[\frac{1}{4}, \frac{3}{4}\right]$
- Q 138. If $\phi(x) = \frac{1}{1+e^{-x}}$ then the value of
(a) 5 (b) $\frac{9}{2}$ (c) $\frac{11}{2}$ (d) none of these
- Q 139. If $f(x+2) = \frac{1}{2} \left\{ f(x+1) + \frac{4}{f(x)} \right\}$ and $f(x) > 0$ for all $x \in \mathbb{R}$ then $\lim_{x \rightarrow \infty} f(x)$ is
(a) 1 (b) 2 (c) -2 (d) 0
- Q 140. If $I = \int_{1/\pi}^{\pi} \frac{1}{x} \cdot \sin\left(x - \frac{1}{x}\right) dx$ then I is equal to
(a) 0 (b) π (c) $\pi - \frac{1}{\pi}$ (d) $\pi + \frac{1}{\pi}$
- Q 141. If the equations $4x^2 - x - 1 = 0$ and $3x^2 + (\lambda + \mu)x + \lambda - \mu = 0$ have a common root then the rational values of λ and μ are
(a) $\lambda = 0, \mu = -\frac{3}{4}$ (b) $\lambda = -\frac{3}{4}, \mu = 0$ (c) $\lambda = -\frac{3}{4}, \mu = \frac{3}{4}$ (d) $\lambda = -\frac{3}{4}, \mu = \frac{1}{4}$
- Q 142. If $ax^2 + bx + 1 = 0$, $a \in \mathbb{R}$, $b \in \mathbb{R}$, does not have distinct real roots then the least value of $3a - 2b$ is
(a) $-\frac{2}{3}$ (b) -1 (c) $\frac{1}{3}$ (d) $-\frac{4}{3}$
- Q 143. A square is inscribed in the circle $x^2 + y^2 - 2x - 4y = 0$ whose sides are parallel to the axes of reference. A vertex of the square is
(a) (3,1) (b) (-2,3) (c) $\left(\frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}}, \frac{2\sqrt{2}-\sqrt{5}}{\sqrt{2}}\right)$ (d) $\left(\frac{\sqrt{5}+\sqrt{2}}{\sqrt{2}}, -\frac{2\sqrt{2}+\sqrt{5}}{\sqrt{2}}\right)$
- Q 144. Let $f(x) = ax^2 = ax^2 - bx + c^2$, $b \neq 0$ and $f(x) \neq 0$ for all $x \in \mathbb{R}$. Then
(a) $a + c^2 < b$ (b) $4a + c^2 > 2b$ (c) $9a - 3b + c^2 < 0$ (d) none of these
- Q 145. Let $f(3) = 4$ and $f'(3) = 5$. Then $\lim_{x \rightarrow 3} [f(x)]$, where $[.]$ denotes the greatest integer function, is
(a) 3 (b) 4 (c) 5 (d) nonexistent
- Q 146. If $[x + [x]] \leq 2$ where $[x]$ denotes the greatest integer $\leq x$, then x lies in the interval,
(a) $(-\infty, 1]$ (b) $(-\infty, 2)$ (c) $(-\infty, 2]$ (d) $(-\infty, 1)$
- Q 147. Let $A = (3,4)$ and B is a variable point on the lines $|x| = 6$. If $AB \leq 4$ then the number of positions of B with integral coordinates is
(a) 5 (b) 4 (c) 6 (d) 10
- Q 148. Let $A = (3, -4)$, $B = (1, 2)$. $P = (2k - 1, 2k + 1)$ is a variable point such that $PA + PB$ is the minimum. Then k is
(a) $\frac{7}{9}$ (b) 0 (c) $\frac{7}{8}$ (d) none of these

- Q 149. Let $f(x)$ be a function which can be expressed as a power series such that $f(0) = p$, $f'(0) = pq$, $f''(0) = pq^2, \dots, f^n(0) = pq^n, \dots$ where $f^n(0) = \left\{ \frac{d^n f(x)}{dx^n} \right\}_{x=0}$
- Then $\lim_{x \rightarrow p} f(x)$ is equal to
 (a) p (b) q (c) pe^{pq} (d) qe^{pq}
- Q 150. If the angle between the tangents from the point $(\lambda, 1)$ to the parabola $y^2 = 16x$ be $\frac{\pi}{2}$ then λ is
 (a) 4 (b) -4 (c) -1 (d) 2
- Q 151. The number of ways in which 9 flowers, of which 5 are identical and white, and the other four are of different colours, can be set on a garland so that no two of the coloured flowers are consecutive, is
 (a) 60 (b) 120 (c) 180 (d) none of these
- Q 152. If r, r_0 be the inradius and an exradius respectively of an equilateral triangle then $r : r_0$ is equal to
 (a) 1 : 2 (b) 3 : 1 (c) 1 : $\sqrt{3}$ (d) 1 : 3
- Q 153. The value of $\int_0^{\pi/4} \frac{\sec x}{(\sec x + \tan x)^2} dx$ is
 (a) $1 + \sqrt{2}$ (b) $-(1 + \sqrt{2})$ (c) $-\sqrt{2}$ (d) none of these
- Q 154. If the two circles $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points then
 (a) $r < 2$ (b) $r = 2$ (c) $r > 2$ (d) $2 < r < 8$
- Q 155. The number of points with integral coordinates, exactly in the interior of the triangle with vertices $(0, 0)$, $(0, 21)$, $(21, 0)$, is
 (a) 133 (b) 190 (c) 233 (d) 105
- Q 156. If $\lim_{x \rightarrow 0} \frac{\{(a-n)nc - \tan x\} \sin nx}{x^2} = 0$, $n \neq 0$ then a is equal to
 (a) 0 (b) $1 + \frac{1}{n}$ (c) n (d) $n + \frac{1}{n}$
- Q 157. A point on the line $\frac{x-1}{0} = \frac{y}{1} = \frac{z+1}{2}$ at a distance 3 from the plane $x + y + z = 3$ is
 (a) $(1, 1, 1)$ (b) $(1 + \sqrt{3}, 1, 2\sqrt{3} + 1)$ (c) $(1, \sqrt{3} + 1, 2\sqrt{3} + 1)$ (d) $(1 - \sqrt{3}, 1, 2\sqrt{3} - 1)$
- Q 158. Two numbers are selected one by one without replacement at random from the set $S = \{1, 2, 3, 4, 5, 6\}$. The probability that the smaller of the two numbers is less than 4 is
 (a) $\frac{1}{15}$ (b) $\frac{14}{15}$ (c) $\frac{1}{5}$ (d) $\frac{4}{5}$
- Q 159. Given $f'(x) = 6$ and $f(1) = 4$, $\lim_{h \rightarrow 0} \frac{f(2h + 2 + h^2) - f(2)}{f(h - h^2 + 1) - f(1)}$ is
 (a) 3 (b) $-\frac{3}{2}$ (c) $\frac{3}{2}$ (d) does not exist
- Q 160. The domain of the function $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ for real x , is
 (a) $\left[-\frac{1}{4}, \frac{1}{2}\right]$ (b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (c) $\left(-\frac{1}{2}, \frac{1}{9}\right)$ (d) $\left[-\frac{1}{4}, \frac{1}{4}\right]$
- Q 161. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ such that $\min f(x) > \max g(x)$ then the relation between b and c is
 (a) $0 < c < b\sqrt{2}$ (b) $|c| < \sqrt{2} |b|$ (c) $|c| > \sqrt{2} |b|$ (d) no real b and c
- Q 162. The coefficient of x^{24} in $(1 + x^2)^{12} \cdot (1 + x^{12}) \cdot (1 + x^{24})$ is
 (a) ${}^{12}C_6 + 3$ (b) ${}^{12}C_6 + 1$ (c) ${}^{12}C_6$ (d) ${}^{12}C_6 + 2$

- Q 163. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals then $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in
 (a) GP (b) HP (c) AP (d) arithmetico-geometric progression
- Q 164. Let $\vec{u} = \vec{i} + \vec{j}, \vec{v} = \vec{i} - \vec{j}$ and $\vec{w} = \vec{i} + 2\vec{j} + 3\vec{k}$. is a unit vector such that $\vec{u} \cdot \vec{n} = 0$ and $\vec{v} \cdot \vec{n} = 0$ then $|\vec{w} \cdot \vec{n}|$ is equal to
 (a) 1 (b) 2 (c) 3 (d) 0
- Q 165. Let $f'(x) = f(x)$ where $f(0) = 1$. If $f(x) + g(x) = x^2$ then $\int_0^1 f(x)g(x)dx$ is equal to
 (a) $\frac{e^2}{2} + e - \frac{3}{2}$ (b) $-\frac{e^2}{2} + e - \frac{3}{2}$ (c) $\frac{e^2}{2} + e + \frac{5}{2}$ (d) $-\frac{e^2}{2} + e - \frac{5}{2}$
- Q 166. Let a, b, c be distinct real numbers such that $a^2 - b = b^2 - c = c^2 - a$. Then $(a + b)(b + c)(c + a)$ equals
 (a) 0 (b) 1 (c) -1 (d) none of these
- Q 167. The number of elements in the set $S = \{i^n + i^{-n} : n \text{ is an integer}\}$ where $i = \sqrt{-1}$, is
 (a) 3 (b) 1 (c) greater than 4 but finite (d) infinite
- Q 168. If $a_0 = 1, a_1 = 1$ and $a_n = a_{n-1} \cdot a_{n-2} + 1, n > 2$, then
 (a) a_{165} is odd, a_{166} is even (b) a_{165} is even, a_{166} is odd
 (c) both a_{165}, a_{166} are odd (d) both a_{165}, a_{166} are even
- Q 169. If $P(B) = \frac{3}{4}, P(A \cap B \cap \bar{C}) = \frac{1}{3}$ and $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$ then $P(B \cap C)$ is
 (a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{15}$ (d) $\frac{1}{5}$
- Q 170. If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$ then $f(x)$ increases in
 (a) $(0, \infty)$ (b) $(-\infty, 0)$ (c) $(2, 2)$ (d) no value of x
- Q 171. Events A, B, C are mutually exclusive events such that $P(A) = \frac{3x+1}{3}, P(B) = \frac{1-x}{4}$ and $P(C) = \frac{1-2x}{2}$. The set of possible values of x is in the interval
 (a) $\left[\frac{1}{3}, \frac{2}{3}\right]$ (b) $\left[\frac{1}{3}, \frac{13}{3}\right]$ (c) $[0, 1]$ (d) $\left[\frac{1}{3}, \frac{1}{2}\right]$
- Q 172. If in a $\triangle ABC$, $b = 3$ cm, $c = 4$ cm and the length of the perpendicular from A to the side BC is 2 cm then the number of solutions of the triangle is
 (a) 0 (b) 1 (c) 2 (d) 3
- Q 173. The solution set of the equation $[2x] = [x] + 3, 0 \leq x \leq 4$ where $[x]$ = the greatest integer less than or equal to x, is
 (a) $\{3\}$ (b) $\left\{3, \frac{5}{2}\right\}$ (c) $\left[\frac{5}{2}, \frac{7}{2}\right)$ (d) $[2, 3]$
- Q 174. The ratio in which the line segment joining the points whose position vectors are $2\vec{i} - 4\vec{j} - 7\vec{k}$ and $2\vec{i} - 4\vec{j} - 7\vec{k}$ is divided by the plane whose equation is $\vec{r} \cdot (\vec{i} - 2\vec{j} - 3\vec{k}) = 13$, is
 (a) 13:12 internally (b) 12 : 25 externally (c) 13:25 internally (d) 37:25 internally
- Q 175. The value of $\sum_{r=0}^n {}^{n+r}C_r$ is equal to
 (a) ${}^{2n+1}C_n$ (b) ${}^{2n}C_{n-1}$ (c) ${}^{2n}C_{n+1}$ (d) ${}^{2n+1}C_{n-1}$
- Q 176. The value of p for which one root of the quadratic. equation $(p^2 - 5p + 3)x^2 + (3p - 1)x + 2 = 0$ is twice as large as the other,
- Q 177. If $f(a + b - x) = f(x)$ then $\int_a^b xf(x)dx$ is equal to

- (a) $\frac{a+b}{2} \int_a^b f(x) dx$ (b) $\frac{b-a}{2} \int_a^b f(x) dx$ (c) $\frac{a+b}{2} \int_a^b f(a+b-x) dx$ (d) $\frac{a+b}{2} \int_a^b f(b-x) dx$
- Q 178. The number of integral terms in the expansion of $(\sqrt{3} + \sqrt[5]{5})^{256}$ is
 (a) 25 (b) 26 (c) 24 (d) none of these
- Q 179. ${}^nC_{n+1} + {}^nC_{r-1} + 2 \times {}^nC_r$ is equal to
 (a) ${}^{n+2}C_{r+1}$ (b) ${}^{n+1}C_r$ (c) ${}^{n+1}C_{r+1}$ (d) ${}^{n+2}C_r$
- Q 180. A focal chord of the parabola $y^2 = 16x$ touches the circle $(x-6)^2 + y^2 = 2$. Then the possible values of the slope of the chord are
 (a) $\{-1, 1\}$ (b) $\{-2, 2\}$ (c) $\left\{-2, \frac{1}{2}\right\}$ (d) $\left\{2, -\frac{1}{2}\right\}$
- Q 181. The value of X so that the volume of the parallelepiped formed by the vectors and becomes minimum, is
 (a) -3 (b) 3 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$
- Q 182. The sum of the radii of inscribed and circumscribed circles for an n-sided regular polygon of side a, is
 (a) $\frac{a}{2} \cot \frac{\pi}{2n}$ (b) $a \cot \frac{\pi}{2n}$ (c) $\frac{a}{4} \cot \frac{\pi}{2n}$ (d) $a \cot \frac{\pi}{n}$
- Q 183. If $|z| = 1$ and $w = \frac{z-1}{z+1}$ ($z \neq -1$) then $\operatorname{Re}(w)$ is
 (a) 0 (b) $\frac{-1}{|z+1|^2}$ (c) $\frac{1}{|z+1|^2}$ (d) $\frac{\sqrt{2}}{|z+1|^2}$
- Q 184. If z and w are two nonzero complex numbers such that $|zw| = 1$ and $\arg z - \arg w = \frac{\pi}{2}$ then $\bar{z}w$ is equal to
 (a) -1 (b) i (c) -i (d) 1
- Q 185. If the system of equations $x + ay = az + y = ax + z = 0$ has infinite solutions then a is
 (a) -1 (b) 1 (c) 0 (d) no real value
- Q 186. The origin, z_1 and z_2 are the vertices of an equilateral triangle where z_1, z_2 are roots of the equation $z^2 + az + b = 0$. Then
 (a) $a^2 - 2b$ (b) $a^2 = 3b$ (c) $a^2 = 4b$ (d) $a = b$
- Q 187. If $I(m, n) = \int_0^1 t^m (1+t)^n dt$ then $I(m, n)$ is equal to
 (a) $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$ (b) $\frac{n}{m+1} I(m+1, n-1)$
 (c) $\frac{2^n + n}{m+1} I(m+1, n-1)$ (d) $\frac{m}{m+1} I(m+1, n-1)$
- Q 188. Let $f(x) = \sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$, $\alpha \in \left(0, \frac{\pi}{2}\right)$. The value of $f(x)$ is always greater than or equal to
 (a) $2 \tan \alpha$ (b) 1 (c) 2 (d) $\sec^2 \alpha$
- Q 189. A function $f: N \rightarrow Z$ is defined by

$$f(n) = \frac{n-1}{2}, \text{ when } n \text{ is odd} = -\frac{n}{2}, \text{ when } n \text{ is even. The function } f \text{ is}$$
 (a) onto but not one-one (b) onto and one-one
 (c) neither onto nor one-one (d) one-one but not onto
- Q 190. If $f: [0, \infty) \rightarrow [0, \infty)$ and $f(x) = \frac{x}{x+1}$ then f is
 (a) one-one and onto (b) one-one but not onto
 (c) onto but not one-one (d) neither one-one nor onto

- Q 191. In $[0, 1]$ Lagrange's Mean-Value theorem is not applicable to
- (a) $f(x) = \frac{1}{2} - x, x < \frac{1}{2}, x \geq \frac{1}{2}$ (b) $f(x) = \frac{\sin x}{x}, x \neq 0$
 (c) $f(x) = x|x|$ (d) $f(x) = |x|$
- Q 192. If $\vec{u}, \vec{v}, \vec{w}$ are three noncoplanar vectors then equals
- (a) $[\vec{u} \vec{v} \vec{w}]$ (b) $[\vec{u} \vec{w} \vec{v}]$ (c) $3[\vec{u} \vec{v} \vec{w}]$ (d) 0
- Q 193. If the system of linear equations
 $x + 2ay + az = 0$
 $x + 4cy + cz = 0$
 has a nonzero solution then a, b, c
- (a) are in GP (b) are in HP (c) are in AP (d) satisfy $a + 2b + 3c = 0$
- Q 94. If the equation $\lambda x^2 - 2x + 3 = 0$ has positive roots for some real λ then
- (a) $\lambda > 0$ (b) $\lambda > \frac{1}{3}$ (c) $0 \leq \lambda \leq \frac{1}{3}$ (d) $\lambda \leq \frac{1}{3}$
- Q 195. If ${}^{2n}C_6$ is the greatest coefficient in the expansion of $(1 + x)^{2n}$, $n \in \mathbb{N}$, then
- (a) 6 (b) 7 (c) 5 (d) 8
- Q 196. The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if k is
- (a) 1 or -1 (b) 0 or -3 (c) 3 or -3 (d) 0 or -1
- Q 197. The equation of the plane bisecting the obtuse angle between the planes $x + y + z = 1$ and $x + 2y - 4z = 5$ is
- (a) $(\sqrt{7} - 1)x + (\sqrt{7} - 2)y + (\sqrt{7} + 4)z + 5 - \sqrt{7} = 0$
 (b) $(\sqrt{7} + 1)x + (\sqrt{7} + 2)y + (\sqrt{7} + 4)z + 5 - \sqrt{7} = 0$
 (c) $(-\sqrt{7} + 1)x + (\sqrt{7} + 2)y + (\sqrt{7} - 4)z = 5 + \sqrt{7}$
 (d) none of these
- Q 198. Two gas companies G_1 and G_2 , situated at $(40, 0)$ and $(0, 30)$ respectively (unit of distance = 1 km), offer to install equally priced gas ovens at a buyer's house. The company G_1 adds a charge of Rs 40/km of aerial distance between the company and the buyer's house while it is Rs 60 in case of the company G_2 . Then the region where it is cheaper to install the oven from the gas company G_1 is
- (a) the interior of the circle $(x + 32)^2 + (y - 54)^2 = 3600$
 (b) the interior of the circle $(x - 54)^2 + (y + 32)^2 = 3600$
 (c) the exterior of the circle $(x + 32)^2 + (y - 54)^2 = 3600$
 (d) the exterior of the circle $(x - 54)^2 + (y + 32)^2 = 3600$

Answers

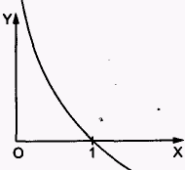
1a	2a	3a	4b	5b	6d	7b	8a	9b	10b
11c	12c	13c	14d	15d	16c	17a	18b	19a	20b
21c	22a	23a	24b	25a	26b	27c	28a	29d	30d
31b	32c	33a	34c	35b	36d	37c	38b	39d	40c
41a	42b	43c	44a	45c	46a	47b	48d	49c	50b
51d	52a	53d	54a	55c	56d	57b	58c	59a	60d
61c	62a	63c	64a	65b	66a	67b	68c	69a	70c
71a	72c	73b	74c	75b	76a	77b	78c	79a	80b
81b	82a	83a	84b	85b	86a	87b	88a	89b	90b
91a	92d	93c	94a	95b	96b	97a	98c	99a	100a
101a	102b	103c	104c	105a	106c	107a	108a	109b	110b
111a	112b	113c	114b	115c	116d	117a	118c	119a	120a
121c	122d	123c	124b	125a	126b	127c	128a	129b	130a
131d	132a	133c	134a	135c	136b	137d	138c	139b	140a
141b	142b	143d	144b	145d	146b	147a	148c	149c	150b
151d	152d	153d	154d	155b	156d	157c	158d	159a	160a
161c	162d	163b	164c	165b	166b	167a	168c	169a	170b
171d	172c	173c	174b	175a	176d	177a	178b	179a	180a
181c	182a	183a	184c	185a	186b	187a	188a	189b	190b
191a	192a	193b	194c	195a	196b	197c	198c		

Miscellaneous Questions

Type – 1

Choose the correct options. One or more options may be correct.

- Q 1. If in a $\triangle ABC$, $\begin{vmatrix} \sin A & \sin B & \sin C \\ \sin B & \sin C & \sin A \\ \sin C & \sin A & \sin B \end{vmatrix} = 0$ then
- (a) $\triangle ABC$ is equilateral (b) $a^3 + b^3 + c^3 = 3abc$ (c) $\sum \sin A = 0$ (d) $\sum \sin^3 A - 3\prod \sin A = 0$
- Q 2. α, β are the roots of $x^2 - px + q = 0$ and α, β, γ are the roots of $x^3 - ax^2 + bx - c = 0$. Then
- (a) $pq + c = aq$ (b) $p + \gamma = a$ (c) $p\gamma + q = b$ (d) $\gamma q = c$
- Q 3. Let $\vec{c} = \lambda \vec{a} + \mu \vec{b} + \gamma \vec{a} \times \vec{b}$ where $\vec{a}$ and $\vec{b}$ are unit vectors such that $\vec{a} \cdot \vec{b} = 0$. Then
- (a) $\mu = \vec{b} \cdot \vec{c}$ (b) $\lambda = |\vec{a} \times \vec{c}|$ (c) $\gamma = [\vec{a} \vec{b} \vec{c}]$ (d) $\lambda + \mu + \gamma = (\vec{a} + \vec{b} + \vec{a} \times \vec{b}) \cdot \vec{c}$
- Q 4. $\triangle ABC$ is an isosceles triangle whose vertices A and C have the coordinates (1, 1) and (2, -1) respectively, and AC is one of the two equal sides. If the equation of the side AB is $x + y = 2$ then B has the coordinates
- (a) (4, -2) (b) (2, 0) (c) $\left(1 - \sqrt{\frac{5}{2}}, 1 + \sqrt{\frac{5}{2}}\right)$ (d) $\left(1 + \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}}\right)$
- Q 5. A vertical, rectangular, thin wall of length = a metres, height = b metres runs from east to west. A vertical flagstaff of height h metres stands on the top edge of the wall, dividing the wall in the ratio $p:q$ ($p > q$). A man starts walking towards the east from the eastern end of the wall. After walking a distance x metres he can just see the top of the flagstaff. Instead, if he walks a distance x metres towards the west from the western end then he can see two-thirds of the flagstaff. Then
- (a) $hx = \frac{3}{4}ab$ (b) $hx = \frac{4}{3}ab$ (c) $p = 2q$ (d) $p = 3q$
- Q 6. Which of the following is/are equal to $(\vec{u} \times \vec{v}) \cdot \vec{w}$?
- (a) $\vec{u} \cdot (\vec{v} \times \vec{w})$ (b) $(\vec{v} \times \vec{w}) \cdot \vec{u}$ (c) $\vec{w} \cdot (\vec{u} \times \vec{v})$ (d) $\vec{v} \cdot (\vec{w} \times \vec{u})$
- Q 7. For the vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ which of the following is/are meaningful if λ and μ represent scalar and vector products respectively?
- (a) $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ (b) $\{\vec{a} \times (\vec{b} \times \vec{c})\} \cdot \vec{d}$ (c) $\vec{a} \times \{(\vec{b} \times \vec{c}) \cdot \vec{d}\}$ (d) $\{(\vec{a} \cdot \vec{b}) \times \vec{c}\} \cdot \vec{d}$
- Q 8. If a, b, c are three consecutive integers then the determinant
- $$\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix}$$
- is divisible by
- (a) 36 (b) 144 (c) 48 (d) none of these
- Q 9. Let P, A, B, C be four collinear points in order, the distances of A, B, C from P being a, b, c respectively. If the equation
- $$(b - a)x^2 + (a - c)x + c - b = 0$$
- has one root double the other then
- (a) B divides AC in the ratio 2 : 1 internally
 (b) C divides AB in the ratio 2 : 1 externally
 (c) B divides AC in the ratio 1 : 2 internally
 (d) none of these
- Q 10. If $x^2 + px + q$ is an integer for all integral values of x then
- (a) p is an integer (b) q is an integer (c) $\frac{p+q}{2}$ is an integer (d) $\frac{p-q}{2}$ is an integer

- Q 11. The equation $px^3 - (p + 1)x^2 + px = 0$ has real roots and p is any positive integer. Then the equation has
 (a) exactly one root less than 1 (b) more than one root < 1
 (c) all roots greater than 1 (d) none of these
- Q 12. If $4x^2 + x + 12k^2 \geq 47$ for all $x \in \mathbb{R}$ then λ can have the value
 (a) 1 (b) 2 (c) -2 (d) 1
- Q 13. Let $f(x) = 5a^2 + 4(x - 1) - x^2$, $x \in \mathbb{R}$ and a is a real constant. If the maximum value of $f(x)$ is 20 then a is
 (a) -2 (b) 4 (c) 2 (d) none of these
- Q 14. If $A = \int_0^1 \sin^{-1} x \, dx$ and $B = \int_0^1 \cos^{-1} x \, dx$ then $\frac{\pi}{2}$ is equal to
 (a) $A - B + 2$ (b) $A - 2B$ (c) $A + 1$ (d) $A + B$
- Q 15. The equation of a parabola is $x + y = 0$. The line $x = 1$, after getting reflected from the parabola, passes
 (a) along the line $3x + 4y + 1 = 0$ (b) through the point $(-1, 1)$
 (c) through the point $(0, -\frac{1}{4})$ (d) through the point $(\frac{1}{3}, \frac{-1}{2})$
- Q 16. If $f(x) = \{x - [x]\} \cos \pi x$ where $[\cdot]$ is the greatest integer function then $f(x)$ is
 (a) an even function (b) an odd function (c) a periodic function (d) a many-one function
- Q 17. Let $f(x)$ be a continuous even periodic function of period a and $f(0) = 0$. Then
 (a) $f(x)$ is monotone in $[0, \frac{a}{2}]$ (b) $f(\frac{a}{2} - x) = f(\frac{a}{2} + x)$ for all x
 (c) $f(x)$ is differentiable in $[0, \frac{a}{2}]$ (d) $f(x)$ has a maximum value
- Q 18. If in a $\triangle ABC$, $a = \sqrt{3}$, $c = 3$ and $A = 30^\circ$ then the angle B has the measure
 (a) 90° (b) 60° (c) 30° (d) 120°
- Q 19. There are $n + 1$ identical balls on which the numbers ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are written, each ball carrying a different number. The total number of ways in which the balls can be arranged in a row so that the values of numbers on the first and the last are equal, is
 (a) $n!$ when n is even (b) $(n^2 - 1) \cdot (n - 2)!$ when n is odd
 (c) $n!$ when n is even (d) $(n + 1) \cdot (n - 1)!$ when n is odd
- Q 20. The figure given below can be the rough sketch of the function given by

- (a) $y = \log_2 x$ (b) $x \cdot 2^y = 1$ (c) $x = \log_{0.5} y$ (d) $y = \log_{0.5} x$
- Q 21. If $px + qx + r = 0$ has no real roots for real values of p, q, r and $4p + 2q + r > 0$ then
 (a) $r > 0$ (b) $p + q + r > 0$ (c) $p + q + r < 0$ (d) $r > 0$
- Q 22. If $f(x) = 3x^2 + 12x - 1$, $x \in [-1, 2]$
 $37 - x$, $x \in (2, 3]$ then
 (a) $f(x)$ is monotonic increasing in $[-1, 2]$ (b) $f(x)$ is continuous in $[-1, 3]$
 (c) $f(x)$ has the maximum value at $x = 2$ (d) $f'(2)$ does not exist
- Q 23. P and Q are two points on the curve $x^2 + y^2 = a^2$, where $\angle POQ = \frac{\pi}{2}$ O being the origin. The equation of the tangent to the curve at P is $x - \sqrt{3}y + 2a = 0$. The equation of the tangent to the curve at Q is
 (a) $x + \sqrt{3}y - 2a = 0$ (b) $\sqrt{3}x + y - 2a = 0$ (c) $x + 2y - 2a = 0$ (d) $\sqrt{3}x + y + 2a = 0$

Answers

1abc	2abcd	3acd	4ac	5ad	6abc	7ab	8abc	9ac	10ab
11ab	12bc	13ac	14acd	15acd	16cd	17bd	18ac	19ab	20bd
21ab	22abcd	23bd							

Miscellaneous Questions

Type – 1

Choose the most appropriate option (a, b, c or d).

- Q 1. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \frac{3}{n^2} \sec \frac{9}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$ equals
 (a) $\frac{1}{2} \operatorname{cosec} 1$ (b) $\frac{1}{2} \sec 1$ (c) $\frac{1}{2} \tan 1$ (d) $\tan 1$
- Q 2. If A be a 3×3 matrix and I be the unit matrix of that order such that $A = A^2 + I$ then A^{-1} is equal to
 (a) A (b) $A + I$ (c) $1 - A$ (d) $A - I$
- Q 3. If $w = \frac{z}{z - 1/3i}$ and $|w| = 1$ then z lies on
 (a) a circle (b) an ellipse (c) a parabola (d) a straight line
- Q 4. The value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} - \dots + \binom{30}{20}\binom{30}{30}$ is,
 (a) $\binom{30}{10}$ (b) $\binom{30}{15}$ (c) $\binom{60}{30}$ (d) $\binom{31}{10}$
- Q 5. If a, b, c are in AP and x, y, z are finite numbers such that $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty}$ and $z = \sum_{n=0}^{\infty} e^n$ then x, y, z are in
 (a) AP (b) GP (c) HP (d) Arithmetio-geometric progression
- Q 6. In a triangle ABC, $a^2 + b^2 = c^2$. If R and r its circumradius and inradius respectively then $R + r$ is equal to
 (a) AM of a and b (b) AM of b and c (c) AM of c and a (d) perimeter
- Q 7. Suppose f(x) is differentiable at $x = 1$ and $\lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$. Then $f'(1)$ equals
 (a) 4 (b) 3 (c) 6 (d) 5
- Q 8. If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$ and $f(0) = 0$ then $f(1)$ equals
 (a) 0 (b) -1 (c) 1 (d) 2
- Q 9. If $\int_{\sin x}^1 t^2 \cdot f(t) dt = 1 - \sin x$ then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to
 (a) $\frac{1}{3}$ (b) -3 (c) 3 (d) $\sqrt{3}$
- Q 10. If f(x) is a twice differentiable function for which $f(1) = 1, f(2) = 4$ and $f(3) = 9$ then
 (a) $f''(x) = 2$ for all $x \in (1, 3)$ (b) $f''(x) = f'(x) = 5$ for some $x \in (2, 3)$
 (c) $f''(x) = 3$ for all $x \in (2, 3)$ (d) $f''(x) = 2$ for some $x \in (1, 3)$
- Q 11. Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$ then the value of $f(6)$ is
 (a) less than 8 (b) greater than or equal to 8
 (c) equal to 5 (d) less than 5
- Q 12. The value of $\int_1^e \left| \frac{\log x - 1}{1 + (\log x)^2} \right| dx$ is equal to
 (a) $\frac{1}{2}(e - 2)$ (b) $-\frac{(e - 1)^2}{2(1 + e^2)}$ (c) $\frac{1}{2}(e - 1)$ (d) $\frac{1}{2}e$
- Q 13. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$ where $\overline{A}$ = complementary of event A. Then A and B are
 (a) equally likely but not independent (b) equally likely and mutually exclusive

(c) mutually exclusive and independent (d) independent but not equally likely

- Q 14. Let $\vec{a} = \vec{i} - \vec{k}$, $\vec{b} = x\vec{i} + \vec{j} + (1-x)\vec{k}$ and $\vec{c} = y\vec{i} + x\vec{j} + (1-x-y)\vec{k}$. Then $[\vec{a} \vec{b} \vec{c}]$ depends on
 (a) only x (b) only y (c) neither x nor y (d) both x and y

- Q 15. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three nonzero noncoplanar vectors and $\vec{p} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$, $\vec{q} = \vec{b} + \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$,

$$\vec{r} = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} + \frac{\vec{b} \cdot \vec{c}}{|\vec{c}|^2} \vec{p}, \quad \vec{s} = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{p} \cdot \vec{c}}{|\vec{p}|^2} \vec{p},$$

$$\vec{t} = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} + \frac{\vec{b} \cdot \vec{c}}{|\vec{c}|^2} \vec{p}, \quad \vec{u} = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{b} \cdot \vec{c}}{|\vec{b}|^2} \vec{p}$$

Then the set of orthogonal vectors is

- (a) $(\vec{a}, \vec{p}, \vec{t})$ (b) $(\vec{a}, \vec{p}, \vec{s})$ (c) $(\vec{a}, \vec{p}, \vec{u})$ (d) $(\vec{a}, \vec{q}, \vec{s})$
 Q 16. A six faced fair dice is thrown until 1 comes. The probability that 1 comes in even number of trials, is

- (a) $\frac{5}{11}$ (b) $\frac{5}{6}$ (c) $\frac{6}{11}$ (d) $\frac{1}{6}$

- Q 17. Four flats are available Krishna Apartment. Four persons visit the flats and each gives his choice to the promoter without consulting others. The probability that all will give different choice, is

- (a) $\frac{1}{8}$ (b) $\frac{1}{64}$ (c) $\frac{3}{32}$ (d) $\frac{3}{8}$

- Q 18. The area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

- (a) ab (b) $2ab$ (c) $\frac{a}{b}$ (d) $\sqrt{ab}$

- Q 19. The value of $(\vec{a} \times \vec{i})^2 + (\vec{a} \times \vec{j})^2 + (\vec{a} \times \vec{k})^2$ is equal to

- (a) $\vec{a}^2$ (b) $3\vec{a}^2$ (c) $4\vec{a}^2$ (d) $2\vec{a}^2$

- Q 20. The minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the coordinate axes, is

- (a) ab (b) $\frac{a^2 + b^2}{2}$ (c) $\frac{(a+b)^2}{2}$ (d) $\frac{a^2 + ab + b^2}{3}$

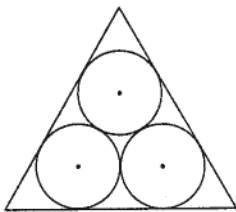
- Q 21. If y is a function of x given by $ydx + y^2dy = xdy$ where $y > 0$ and $y(1) = 1$ then $y(-3)$ is

- (a) 3 (b) 2 (c) 1 (d) 5

- Q 22. The function $y = ||x| - 1|$ is differentiable for all $x \in \mathbb{R}$ except for the values

- (a) 0, 1, -1 (b) ± 1 (c) 1 (d) -1

- Q 23. 3 coins radii 1 unit are kept inside and equilateral triangle so that they touch each other and also the sides of the triangle. The area of the triangle is

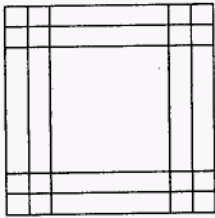


- (a) $(4 + 2\sqrt{3}) \text{ unit}^2$ (b) $(6 + 4\sqrt{3}) \text{ unit}^2$ (c) $\left(12 + \frac{7\sqrt{3}}{4}\right) \text{ unit}^2$ (d) $\left(3 + \frac{7\sqrt{3}}{4}\right) \text{ unit}^2$
- Q 24. If a, b, c are integers not all equal and $w = \sqrt[3]{1}$, ($w \neq 1$) then the minimum value of $|a + bw + cw^2|$ is
 (a) 0 (b) 1 (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$
- Q 25. If both the roots of the quadratic equation $x^2 - 2kx + k - 5 = 0$ are less than 5 then k lies in the interval
 (a) $(6, \infty)$ (b) $(5, 6]$ (c) $[4, 5]$ (d) $(-\infty, 4)$
- Q 26. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in GP then the determinant

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$$

 is equal to
 (a) 0 (b) 1 (c) 2 (d) 4
- Q 27. A real valued function $f(x)$ satisfies the functional equation $f(x - y) = f(x)f(y) - (a - x)(a + y)$ where a is a given constant and $f(0) = 1$. Then $f(2a - x)$ equals
 (a) $f(x)$ (b) $-f(x)$ (c) $f(-x)$ (d) $f(a) + f(a - x)$
- Q 28. Let $f: (-1, 1) \rightarrow B$ be a function defined by $f(x) = \tan^{-1} \frac{2x}{1-x^2}$. Then f is both one-one and onto function when B is the interval
 (a) $\left(0, \frac{\pi}{2}\right]$ (b) $\left(0, \frac{\pi}{2}\right)$ (c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- Q 29. The value of a for which the sum of the squares of the roots of the equation $x^2 - (a-2)x - a - 1 = 0$ assume the least value is
 (a) 0 (b) 1 (c) 2 (d) 3
- Q 30. The system of equations $\alpha x + y + z = \alpha - 1$, $x + \alpha y + z = \alpha - 1$, $x + y + \alpha z = \alpha - 1$ has no solution if α is
 (a) either -2 or 1 (b) -2 (c) 1 (d) 2
- Q 31. A circle is given by $x + (y - 1) = 1$. Then the locus of the centre of the circle touching the given circle and the x -axis, is
 (a) $\{(x, y) | x^2 = 4y\} \cup \{(x, y) | y \leq 0\}$ (b) $\{(x, y) | x^2 + (y - 1)^2 = 4\} \cup \{(x, y) | y \leq 0\}$
 (c) $\{(x, y) | x^2 = 4y\}$ (d) $\{(x, y) | x^2 = 4y\} \cup \{(0, y) | y \leq 0\}$
- Q 32. Let $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = \frac{1}{e}$, where $\alpha, \beta \in [-\pi, \pi]$ and e is the exponential number. The number of values of the pair (α, β) satisfying both the equations, is
 (a) 1 (b) 2 (c) 3 (d) 4
- Q 33. The value of $\int_{-2}^0 \{x^3 + 3x^2 + 3x + 3 + (x + 1)\cos(x + 1)\} dx$ is $J \{x + 3x^* + 3x + 3 + (x + 1)\cos(x + 1)\} dx$ is
 (a) -4 (b) 0 (c) 4 (d) 6
- Q 34. Let α, β be roots of the equation $ax^2 + bx + c = 0$ and $\Delta = b^2 - 4ac$. If $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in GP then
 (a) $\Delta \neq 0$ (b) $\Delta = 0$ (c) $\Delta = 0$ (d) $\Delta = 0$
- Q 35. The number of measures of the angle A for which $\cos A, \cos 2A$ and $\cos 3A$ are in GP, is
 (a) 0 (b) 2 (c) 4 (d) infinite
- Q 36. If z_1 and z_2 are two nonzero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$ then $\arg z_1 - \arg z_2$ is equal to
 (a) $-\pi$ (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{2}$ (d) 0

- Q 37. If $a^2 + b^2 + c^2 + 2 = 0$ and $f(x) = \begin{vmatrix} 1+a^2x & (a+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then $f(x)$ is a polynomial of degree
- (a) 0 (b) 1 (c) 2 (d) 3
- Q 38. Let a and B be the distinct roots of $ax^2 + bx + c = 0$ then is equal to
- $$\lim_{n \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$$
- is equal to
- (a) 0 (b) $\frac{1}{2} a^2(\alpha - \beta)^2$ (c) $\frac{1}{2} (\alpha - \beta)^2$ (d) $-\frac{1}{2} a^2(\alpha - \beta)^2$
- Q 39. If $I_1 = \int_0^1 2^{x^2} dx, I_2 = \int_0^1 2^{x^2} dx, I_3 = \int_1^2 dx$ and $I_4 = \int_1^2 2^{x^3} dx$ then
- (a) $I_1 > I_2$ (b) $I_2 > I_1$ (c) $I_1 = I_2$ (d) $I_3 > I_4$
- Q 40. The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is
- (a) $\frac{\pi}{2}$ (b) 0 (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$
- Q 41. If the angle θ between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda} z + 4 = 0$ is such that $\sin \theta = \frac{1}{3}$ then the value of λ is
- (a) $-\frac{3}{5}$ (b) $\frac{5}{3}$ (c) $-\frac{4}{3}$ (d) $\frac{3}{4}$
- Q 42. The tangent to the curve $y = x^2 + 6$ at $(1, 7)$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point
- (a) $(-6, -11)$ (b) $(-9, -13)$ (c) $(-10, -1-5)$ (d) $(-6, -7)$
- Q 43. The distance between the line $\vec{r} = 2\vec{i} - 2\vec{j} + 3\vec{k} + \lambda(\vec{i} - \vec{j} + 4\vec{k})$ and the plane $\vec{r} \cdot (\vec{i} + 5\vec{j} + \vec{k}) = 5$ is
- (a) $\frac{10}{3\sqrt{3}}$ (b) $\frac{10}{9}$ (c) $\frac{10}{3}$ (d) 0
- Q 44. If $f(x)$ is continuous and differentiable function and $f\left(\frac{1}{n}\right) = 0$ for all $n > 1$ and $n \in \mathbb{I}$ then
- (a) $f(x) = 0, x \in (0, 1]$ (b) $f(0) = 0, f'(0) = 0$
(c) $f'(0) = 0 = f''(0), x \in (0, 1]$ (d) $f(0) = 0$ and $f'(0)$ need not be zero
- Q 45. If $f(x)$ and $g(x)$ are real functions defined by
 $f(x) = 0, x$ is rational $g(x) = 0, x$ is irrational
and
 x, x is irrational x, x is rational then $(f - g)(x)$ is
- (a) one-one and onto (b) neither one-one nor onto
(c) one-one but not onto (d) onto but not one-one
- Q 46. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x} dx, a > 0$ is
- (a) $\frac{\pi}{2}$ (b) $a\pi$ (c) 2π (d) $\frac{\pi}{a}$
- Q 47. If the pair of lines $ax^2 + 2(a + b)xy + by^2 = 0$ lie along two diameters of the circle $x^2 + y^2 = 4$ such that the circle is divided into four sectors in which one sector has double the area of another sector then
- (a) $a = 2b$ (b) $2a = b$ (c) $a = b$ (d) $a + b + 1 = 0$
- Q 48. A rectangle with side lengths $1m - 1$ and $2n - 1$ is divided into unit squares by drawing parallel lines as shown in the diagram. Then the number of rectangles possible with odd side lengths is



- (a) $(m+n+1)^2$ (b) 4^{m+n-1} (c) m^2n^2 (d) $mn(m+1)(n+1)$
- Q 49. Let $A = \{x | x \in \mathbb{N}, x \leq 15\}$. If $3^m + 3^n$ is divisible by 5 where $(m, n) \in A \times A$ then the number of possible values of the ordered pair (m, n) is
 (a) 32 (b) 56 (c) 28 (d) 64
- Q 50. If z_1, z_2 be two n th roots of unity such that they represent two points A, B in the Argand plane where $\angle AOB = 60^\circ$ and O is the origin then the positive integer n is of the form
 (a) $4k, k \in \mathbb{N}$ (b) $4k+3, k \in \mathbb{N}$ (c) $6k, k \in \mathbb{N}$ (d) $6k+5, k \in \mathbb{N}$
- Q 51. If the circle passing through the points $(0, 0)$, $(1, 0)$ and $(0, 1)$ also passes through the point $\left(\lambda, \frac{1}{\lambda}\right)$ then λ is
 (a) 1 (b) -1 (c) 2 (d) -2
- Q 52. If in a $\triangle ABC$, $\tan A + \tan B + \tan C$ has the value 9 then the value of $\cot A \cdot \cot B \cdot \cot C$ is equal to
 (a) 9 (b) 1 (c) $\frac{1}{9}$ (d) none of these
- Q 53. The number of negative integral solutions of the equation $x_1 + x_2 + x_3 + x_4 + 15 = 0$ is
 (a) 364 (b) ${}^{15}C_3$ (c) ${}^{14}C_4$ (d) none of these
- Q 54. Which of the following is equal to $\sqrt[3]{-1}$?
 (a) $\frac{\sqrt{3} + \sqrt{-1}}{2}$ (b) $\frac{-\sqrt{3} + \sqrt{-1}}{\sqrt{-4}}$ (c) $\frac{\sqrt{3} - \sqrt{-1}}{\sqrt{-4}}$ (d) $-\sqrt{-1}$
- Q 55. The number of values of a for which $(x^2 + x + 1)a^2 - (x^2 + 2)a - 3(2x^2 + 3x + 1) = 0$ will be an identity in x is
 (a) one (b) zero (c) three (d) two
- Q 56. If α, β are two values of θ satisfying the equation $\sec^2\theta + p\tan\theta + q = 1$ then
 (a) then $(\alpha + \beta) = \frac{p}{q-1}$ (b) $\tan(\alpha - \beta) = \frac{p}{q-1}$ (c) $\tan(\alpha + \beta) = \frac{q}{p-1}$ (d) $\tan(\alpha - \beta) = \frac{q}{p-1}$
- Q 57. The set of values of a for which the graph of the function $f(x) = a(x+1) + x$ is always above the x -axis is
 (a) $\left(\frac{1}{2}, +\infty\right)$ (b) $\left(-\infty, -\frac{1}{2}\right)$ (c) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ (d) $(0, +\infty)$
- Q 58. $\lim_{x \rightarrow \infty} \frac{x^3 - x}{\sqrt{1+x^6}}$ is equal to
 (a) -1 (b) 1 (c) 0 (d) 3
- Q 59. Three particles A, B and C start to move from three vertices of a triangle with the same uniform speed along the sides of the triangle at random. The probability that they will collide is
 (a) $\frac{3}{8}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$
- Q 60. $\{(\vec{a} \times \vec{b}) \times \vec{a}\} \times \vec{b} + \vec{a} \times \{\vec{b} \times (\vec{a} \times \vec{b})\}$ is equal to
 (a) $2(\vec{a} \cdot \vec{b})\vec{b} \times \vec{a}$ (b) $\vec{0}$ (c) $2\vec{a} \times \vec{b}$ (d) $(\vec{a} \times \vec{b})^2$
- Q 61. The four common tangents to the ellipses $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and $\frac{x^2}{4} + \frac{y^2}{9} = 1$ form
 (a) a rectangle of area 13 unit
 (b) a square of area 26 unit
 (c) a parallelogram which is neither a square nor a rectangle
 (d) a rhombus

- Q 62. A bag contains 12 tickets numbered 1, 2, 3, ..., 12. Five of them are drawn at random and arranged in the ascending order of their numbers. What is the probability that the third in the order is 5? -
- (a) $\frac{7}{44}$ (b) $\frac{5}{12}$ (c) $\frac{2}{3}$ (d) $\frac{1}{2}$
- Q 63. The lines $\frac{x}{0} = \frac{y}{1} = \frac{z}{1}, \frac{x}{2} = \frac{y}{5} = \frac{z}{7}$ and $\frac{x}{1} = \frac{y-1}{2} = \frac{z-1}{3}$
- (a) are three concurrent lines
 (b) are three lines of which at least two are skew lines
 (c) are coplanar lines forming a triangle of area $2\sqrt{3}$ unit²
 (d) form a triangle of area $\sqrt{3}$ unit²
- Q 64. If $(1-y)^m \cdot (1+y)^n = 1 + a_1y + a_2y^2 + \dots + a_{m+n}y^{m+n}$, where $m \in \mathbb{N}$, $n \in \mathbb{N}$ and $a_1 = a_2 = 10$, then (m, n) is
- (a) (20, 45) (b) (35, 20) (c) (45, 35) (d) (35, 45)
- Q 65. If (a, a^2) falls inside the angle made by the lines $y = \frac{1}{2}x (x > 0)$ and $y = 3x (x > 0)$ then a belongs to
- (a) $\left(0, \frac{1}{2}\right)$ (b) $(3, +\infty)$ (c) $\left(\frac{1}{2}, 3\right)$ (d) $\left(-3, -\frac{1}{2}\right)$
- Q 66. If $\frac{1}{(1-ax)(1-bx)} = a_0 + a_1x + a_2x^2 + \dots$ then a_n equals
- (a) $\frac{b^n - a^n}{b-a}$ (b) $\frac{a^n - b^n}{b-a}$ (c) $\frac{a^{n+1} - b^{n+1}}{b-a}$ (d) $\frac{b^{n+1} - a^{n+1}}{b-a}$
- Q 67. The locus of vertices of the family of parabolas where a is a parameter, is
- (a) a circle (b) an ellipse (c) a parabola (d) a rectangular hyperbola
- Q 68. Let E^c denote the complement of an event. Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P(E^c \cap F^c | G)$ equals
- (a) $P(E^c) + P(F^c)$ (b) $P(E^c) - P(F^c)$ (c) $P(E^c) - P(F)$ (d) $P(E) - P(F^c)$

Answers

1c	2c	3d	4a	5c	6a	7d	8a	9c	10d
11b	12a	13d	14a	15b	16a	17c	18b	19d	20a
21a	22a	23b	24b	25d	26a	27b	28c	29b	30b
31d	32c	33c	34c	35a	36d	37c	38b	39a	40a
41b	42d	43a	44b	45a	46a	47c	48c	49b	50c
51a	52c	53a	54b	55a	56a	57a	58a	59c	60a
61b	62a	63d	64d	65c	66d	67d	68c		

Miscellaneous Questions

Type – 1

Comprehension-Type Questions

• $2^m 3^n 5^p$ is a divisor of $2^{10} \cdot 3^8 \cdot 5^7$ if m, n and p are all whole numbers such that $0 \leq m \leq 10, 0 \leq n \leq 8$ and $0 \leq p \leq 7$. Also,
 $(2^0 + 2^1 + 2^2 + \dots + 2^{10})(3^0 + 3^1 + 3^2 + \dots + 3^8)(5^0 + 5^1 + 5^2 + \dots + 5^7) = \text{sum of all the divisors of } 2^{10} \cdot 3^8 \cdot 5^7$.

- Q 1. The number of proper divisors of 16200 is
 (a) 60 (b) 24 (c) 58 (d) 22
- Q 2. The sum of proper divisors of 360 is
 (a) 1170 (b) 810 (c) 1169 (d) 809
- Q 3. The sum of odd divisors ($\neq 1$) of 10800 is
 (a) 1239 (b) 1240 (c) 1238 (d) 2479
- We know that the number of positive integral solutions of the equation $x_1 + x_2 + x_3 + \dots + x_m = n$ ($n \in \mathbb{N}$) is equal to the coefficient of x^n in the expansion of $(x + x^2 + x^3 + \dots \text{ to } \infty)^m$ when $|x| < 1$. Also, we have the expansion
 $(1 - x)^{-n} = {}^{n-1}C_0 + {}^nC_1 x + {}^{n+1}C_2 x^2 + \dots + {}^{n+r}C_{r+1} x^{r+1} + \dots \text{ to } \infty$,
 where $|x| < 1$.
- Q 4. The number of positive integral solutions of the equation $x_1 + x_2 + x_3 + x_4 = 10$ is
 (a) 9C_6 (b) ${}^{10}C_4$ (c) 9C_4 (d) ${}^{10}C_5$
- Q 5. The number of negative integral solutions of the equation $x_1 + x_2 + x_3 + x_4 + x_5 + 10 = 0$ is
 (a) ${}^{10}C_4$ (b) ${}^{10}C_5$ (c) 9C_5 (d) 9C_6
- Q 6. The number of ways in which 10 identical things can be distributed among 3 persons so that each gets at least one thing is
 (a) 8C_3 (b) ${}^{10}C_4$ (c) 9C_3 (d) 9C_2

• r is called the rank of the matrix A if there exists at least one nonzero minor of order r and every minor of order $r + 1$ of the matrix equals zero. Again if A is a square matrix and I is the unit matrix of the same order then the equation $|A - xI| = 0$ is called the characteristic equation and the roots of the characteristic equation are called eigenvalues of the matrix A .

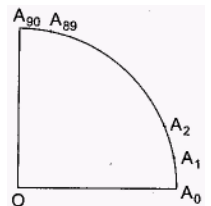
$$\text{Let } A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

- Q 7. The rank of A is
 (a) 3 (b) 2 (c) 1 (d) none of these
- Q 8. The characteristic equation of A is
 (a) $x^3 - 18x^2 + 45x = 0$ (b) $x^3 - 6x^2 + 9x - 4 = 0$ (c) $x^3 + 18x^2 - 45x = 0$ (d) $x^3 + 6x^2 - 9x + 4 = 0$
- Q 9. One of the eigenvalues of A is
 (a) 1 (b) 2 (c) 3 (d) 5

• If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are in an AP whose common difference is λ then we have

$$\sum_{i=1}^n \sin \alpha_i = \frac{\sin \frac{n\lambda}{2}}{\sin \frac{\lambda}{2}} \sin \frac{\alpha_1 + \alpha_n}{2}$$

A quadrant $A_0 A_{90}$ of a unit circle is divided into 90 equal parts by inserting the points $A_1, A_2, A_3, \dots, A_{89}$ on the arc of the circle. Let $\cot 0.5^\circ = a$.



- Q 10. $\sum_{i=1}^{90} \sin \angle(\Delta A_0 O A_i)$ is equal to
 (a) $\frac{a+1}{\sqrt{2}}$ (b) $\frac{a-1}{2\sqrt{2}}$ (c) $\frac{a+1}{2a}$ (d) $\frac{a+1}{2}$

Q 11. $\sum_{i=1}^{90} \ar(\Delta A_0 O A_i)$ is equal to

- (a) $\frac{a+1}{4}$ (b) $\frac{a+1}{2\sqrt{2}}$ (c) $\frac{a-1}{2\sqrt{2}}$ (d) $\frac{a+1}{4a}$

Q 12. $\sum_{i=1}^{90} \sqrt{1 + \cos 2\angle A_0 O A_i}$ is equal to

- (a) $\frac{a+1}{2\sqrt{2}}$ (b) $\frac{a-1}{\sqrt{2}}$ (c) $\frac{a-1}{\sqrt{2}a}$ (d) $\frac{a-1}{2}$

• We know from Gregory's series that

$$\theta = \tan \theta - \frac{1}{3} \tan^3 \theta + \frac{1}{5} \tan^5 \theta - \dots \text{ to } \infty, \quad \text{where } |\theta| < \frac{\pi}{4}$$

Q 13. The sum of the series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ to ∞ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 1 (d) $\frac{\pi}{8}$

Q 14. The sum of the series $\frac{1}{1.3} + \frac{1}{5.7} + \frac{1}{9.11} + \dots$ to ∞ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 1 (d) $\frac{\pi}{8}$

Q 15. The sum of the series $1 - \frac{1}{3.3} + \frac{1}{5.3^2} - \frac{1}{7.3^3} + \dots$ to ∞ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{\sqrt{3}}$ (c) $\frac{\pi}{2\sqrt{3}}$ (d) $\frac{\sqrt{3}\pi}{2}$

• A second-degree curve inscribing the ΔABC , where the sides AB, BC and CA have the equations $L_1 = 0$, $L_2 = 0$ and $L_3 = 0$, respectively, has the equation $L_1 L_2 + \lambda L_2 L_3 + \mu L_3 L_1 = 0$, where λ and μ are parameters. Let the equations of AB, BC and CA be $y = 2x$, $y + 2x = 0$ and $x = 1$, respectively.

Q 16. The circumcentre of the ΔABC is

- (a) $\left(\frac{5}{4}, \frac{5}{8}\right)$ (b) $\left(\frac{5}{2}, 0\right)$ (c) $\left(0, \frac{5}{4}\right)$ (d) $\left(\frac{5}{4}, -\frac{5}{8}\right)$

Q 17. The parabola circumscribing ΔABC and passing through the point (4, 4) has the focus

- (a) (0, 1) (b) (4, 0) (c) (1, 0) (d) (0, 4)

Q 18. The parabola circumscribing ΔABC and passing through the point (4, 4) has the latus rectum

- (a) 4 (b) 1 (c) 16 (d) 8

• The director circle of a circle is the locus of points from which tangents to the circle are at right angles. $C_1, C_2, C_3, \dots$ is a sequence of circles such that C_{r+1} is the director circle of C_r . Let the equation of C_1 be $x^2 + y^2 = 4$.

Q 19. The length of a chord of C_{11} which touches C_{10} is

- (a) $64\sqrt{2}$ (b) 64 (c) $32\sqrt{2}$ (d) 32

Q 20. The distance of the chord of contact of tangents of a point on C_{11} with respect to C_{10} from the centre of C_{10} is

- (a) $64\sqrt{2}$ (b) $32\sqrt{2}$ (c) 64 (d) 32

Q 21. The locus of the poles of the tangents of C_9 with respect to C_{10} is

- (a) $x^2 + y^2 = 2 \times 32^2$ (b) $x^2 + y^2 = 32^2$ (c) $x^2 + y^2 = 64^2$ (d) $x^2 + y^2 = 2 \times 16^2$

• The circle drawn on the major axis of an ellipse as a diameter is the auxiliary circle of the

ellipse. Let an ellipse have the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let p be a point on the ellipse and the

perpendicular to the major axis from the point P when produced towards P meets the auxiliary circle at Q.

Q 22. The equation of the locus of the middle point R of PQ is

(a) $\frac{x^2}{a^2} + \frac{9y^2}{(a+b)^2} = 1$ (b) $\frac{x^2}{4a^2} + \frac{4y^2}{b^2} = 1$ (c) $\frac{x^2}{a^2} + \frac{4y^2}{(a+b)^2} = 1$ (d) $\frac{x^2}{a^2} + \frac{4y^2}{(a+b)^2} = \frac{1}{4}$

Q 23. The area enclosed by the locus of R is

(a) $\frac{\pi}{2}a(a+b)$ (b) $2\pi ab$ (c) $\frac{\pi}{3}a(a+b)$ (d) $\frac{\pi}{4}a(a+b)$

Q 24. The point of intersection of the tangents at P and Q (P and Q being different point) to the corresponding curves lies on

(a) $b^2x^2 + a^2y^2 + a^2y^2 - 2b^2x = a^2$ (b) $y = 0$
(c) $b^2x^2 - a^2y^2 + 2b^2x = a^2 - b^2$ (d) $x = 0$

• Two diameters AB and CD of an ellipse are called conjugate diameters if chords parallel to one of them are bisected by the other diameter. It is found that the difference of eccentric angles of A and C is $\frac{\pi}{2}$. Let the equation of the ellipse be $9x^2 + 16y^2 = 144$.

Q 25. If $A = \left(2, \frac{3\sqrt{3}}{2}\right)$ then coordinates of C are

(a) $\left(-2, \frac{3\sqrt{3}}{2}\right)$ (b) $\left(\sqrt{2} - \sqrt{6}, \frac{3(\sqrt{3}+1)}{2\sqrt{2}}\right)$ (c) $\left(-2\sqrt{3}, \frac{3}{2}\right)$ (d) (0, 3)

Q 26. If the equation of the diameter AB is $x = y$ then the equation of the conjugate diameter CD will be

(a) $9x + 16y = 0$ (b) $x + y = 0$ (c) $16x + 9y = 0$ (d) $9x + 16y = 7$

Q 27. The area of the quadrilateral formed by the tangents at the ends of conjugate diameters AB and CD, where $A = \left(\frac{12}{5}, \frac{12}{5}\right)$ is

(a) 96 (b) 24 (c) 12 (d) 48

• The quadratic equation $ax^2 + bx + c = 0$ will have two roots equal to zero if $b = 0$, $c = 0$. Also, roots of $cx^2 + bx + a = 0$ are reciprocal to the roots of $ax^2 + bx + c = 0$.

A straight line $y = mx + c$ will be an asymptote to the curve $f(x, y) = 0$ if the line touches the curve at infinity. As a result, $f(x, mx + c) = 0$ should have two infinite roots.

Q 28. The value of λ for which the equation $\lambda^2 - 4\lambda + 3 + (\lambda^2 + \lambda - 2)x + 6x^2 - 5x^3 = 0$ will have two roots equal to zero is

(a) 3 (b) -2 (c) 0 (d) 1

Q 29. The value of (λ, μ) for which $(\lambda - \mu)x^3 - (\lambda - 2)x^2 - 3x + 7 = 0$ will have two infinite roots is

(a) (0, 0) (b) (2, 2) (c) (1, 1) (d) (2, 0)

Q 30. The equations of asymptotes to the curve $x + 4xy + 3y^2 + 4x - 3y + 1 = 0$ are

(a) $2x + 2y = 7$ and $x + 3y + 15 = 0$ (b) $x + y - 7 = 0$ and $x + 3y + 15 = 0$
(c) $2(x + y) - 7 = 0$ and $2(x + 3y) + 15 = 0$ (d) $x + y = 7$ and $2(x + 3y) + 15 = 0$

• Perpendiculars AP, AQ and AR are drawn to the x-, y- and z-axes, respectively, from the point $A(1, -1, 2)$.

Q 31. The AM of AP^2 , AQ^2 and AR^2 is

(a) 4 (b) 5 (c) 3 (d) 2

Q 32. The equation of the plane PQR is

(a) $2x + 2y - z = 2$ (b) $2x - 2y + z = 2$ (c) $2x - y + z = 2$ (d) $x - 2y + z = 2$

Q 33. The volume of the tetrahedron (A, PQR) is

(a) 4 (b) $\frac{4}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

• A ray of light emanating from the point source $P(1, -3, 2)$ and travelling parallel to the lines

$\frac{x-2}{1} = \frac{y}{3} = \frac{z+1}{2}$ is incident on the plane $x + y - 3z = 0$ at the point Q. After reflecting from the

plane the ray travels along the line QR. It is also known that the incident ray, reflected ray and the normal to the plane at the point of incidence are in the same plane.

Q 34. The point Q is

(a) (3, 15, 6) (b) (3, 6, 3) (c) (-3, -6, -3) (d) (-3, -15, -6)

Q 35. The equations of the line QR are

$$(a) \frac{x-12}{15} = \frac{y-22}{37} = \frac{z-4}{10}$$

$$(b) \frac{x-3}{3} = \frac{y-15}{7} = \frac{z-6}{2}$$

$$(c) \frac{x-3}{15} = \frac{y-6}{37} = \frac{z-3}{10}$$

$$(d) \frac{x+3}{3} = \frac{y+6}{7} = \frac{z+3}{2}$$

Q 36. The equation of the plane PQR is

$$(a) 5x + 2y - z + 3 = 0 \quad (b) 11x - 5y + 2z = 30 \quad (c) 5x - y - z = 6 \quad (d) x - y + z = 6$$

• The sphere whose centre = (α, β, γ) and radius = a , has the equations $(x - \alpha)^2 + (y - \beta)^2 + (z - \gamma)^2 = a^2$. Again, the intersection of a sphere by a plane is a circle.

Q 37. The distance of the centre of the sphere $x^2 + y^2 + z^2 - 2z - 4y = 0$ from the origin is

$$(a) 5 \quad (b) \sqrt{5} \quad (c) 2\sqrt{5} \quad (d) \frac{5}{2}$$

Q 38. The radius of the circle of intersection of the sphere $x^2 + y^2 + z^2 = 9$ by the plane $3x + 4y + 5z = 5$ is

$$(a) \sqrt{\frac{17}{2}} \quad (b) 3 \quad (c) \sqrt{34} \quad (d) \frac{1}{\sqrt{2}}$$

Q 39. The area of the circle on the sphere $x^2 + y^2 + z^2 = 25$ whose centre is $(1, 1, 1)$ is

$$(a) 13\pi \quad (b) 3\pi \quad (c) 22\pi \quad (d) 25\pi$$

• Two one-parameter families of curves are called orthogonal trajectories of one another if each member of a family cuts all the members of the other family at right angles.

It is found that the orthogonal trajectories of the family of curves $f\left(x, y, \frac{dy}{dx}\right) = 0$ is the family of

curves $f\left(x, y, -\frac{dy}{dx}\right) = 0$.

Q 40. The family of parabolas with a common vertex at the origin whose foci are on the x-axis and the directrices are parallel to the y-axis can have the equation (a being a parameter)

$$(a) y^2 = 3ax \quad (b) x^2 = 4ay \quad (c) y^2 = 4a(x + 2) \quad (d) (y - 1)^2 = 4ax$$

Q 41. The orthogonal trajectory of the family of parabolas $y^2 = 4ax$ is

$$(a) x^2 + y^2 = c^2 \quad (b) x^2 + 2y^2 = c^2 \quad (c) 2x^2 + y^2 = c^2 \quad (d) y^2 - x^2 = c^2$$

Q 42. The curve passing through the point $(1, 2)$ that cuts each member of the family of parabolas $y^2 = 4ax$ orthogonally is

$$(a) 2x^2 + y^2 = 6 \quad (b) x^2 + y^2 = 5 \quad (c) x^2 + 2y^2 = 9 \quad (d) y^2 - x^2 = 3$$

• If any point of the closed interval $[\alpha, \beta]$ is a sample point and any point of the closed interval $[a, b] \subseteq [\alpha, \beta]$ be a favorable point for the event E then the probability of the event E is defined by

$$P(E) = \frac{\text{length of } [a, b]}{\text{length of } [\alpha, \beta]}$$

Q 43. The set of values of $p \in \mathbb{R}$ for which the equation $x^2 + px + \frac{1}{4}(p + 2) \geq 0$ for all $x \in \mathbb{R}$ is

$$(a) (-2, 1) \quad (b) (2, +\infty) \quad (c) [-1, 2] \quad (d) [1, 2]$$

Q 44. The set of values of $p \in \mathbb{R}$ for which the equations $x^2 + px + \frac{1}{4}(p + 2) = 0$ will have real roots is

$$(a) [2, +\infty) \quad (b) (2, +\infty) \quad (c) (-\infty, 2) \quad (d) \mathbb{R} - (-1, 2)$$

Q 45. If p is chosen at random from the interval $[0, 6]$ then the probability that the roots of the equation

$$x^2 + px + \frac{1}{4}(p + 2) = 0 \text{ will be real is}$$

$$(a) \frac{3}{5} \quad (b) \frac{1}{2} \quad (c) \frac{3}{4} \quad (d) \frac{2}{3}$$

• In a series of n independent trials for an event of constant probability p , the most probable number r of successes is given by $(n + 1)p - 1 < r < (n + 1)p$. Hence, the most probable number of successes is the integral part of $(n + 1)p$. But $(n + 1)p$ is an integer, the chance of r successes is equal to that of $r + 1$ successes and both $r, r + 1$ are most probable numbers of successes.

- Q 46. A bag contains 2 white balls and 1 black ball. A ball is drawn at random and returned to the bag. The experiment is done 10 times. The probability that a white ball is drawn exactly 5 times is where $a_0, a_1, \dots, a_n$ are digits whose values range from 0 to $r - 1$. Thus, in the scale of 7, the digits will be 0, 1, 2, 3, 4, 5, 6.

Conversion of a denary number to the scale of 5

$$\begin{array}{r} 5 \overline{) 268} \\ 5 \overline{) 53} \dots 3 \\ 5 \overline{) 10} \dots 3 \\ 2 \dots 0 \end{array} \quad \therefore 268 = 2033 \text{ in the scale of 5}$$

Addition of numbers in the scale of 5

$$\begin{array}{r} 34 \\ + 22 \\ \hline 111 \end{array} \quad \begin{array}{l} 34 = 3 \times 5 + 4 \\ 22 = 2 \times 5 + 2 \\ \hline \text{sum} = 5 \times 5 + 6 = 1 \times 5^2 + 1 \times 5 + 1 \end{array}$$

Multiplication of numbers in the scale of 5

$$\begin{array}{r} 23 \\ \times 4 \\ \hline 202 \end{array} \quad \begin{array}{l} 3 \times 4 = 12 = 2 \times 5 + 2 \\ 2 \times 4 + 2 = 10 = 2 \times 5 + 0 \end{array}$$

On the basis of the above discussions, answer the following.

- Q 55. The denary number 43125 in the scale of 6 will be represented by
 (a) 353135 (b) 531353 (c) 515313 (d) 55453
- Q 56. The sum of the numbers 2053 and 412 in the scale of seven is
 (a) 2465 (b) 3020 (c) 3004 (d) none of these
- Q 57. The product of the numbers 5623 and 6 in the scale of eight is
 (a) 41672 (b) 33738 (c) 42562 (d) 45262
- Z is the set of integers and N is the set of natural numbers. If $b \in Z$, $c \in Z$ and $n \in N$ are such that when b and c are divided by n , we get the same remainder then b and c are said to be congruent with respect to modulo n . Symbolically it is written as $b \equiv c \pmod{n}$
- Q 58. Which of the following is true?
 (a) $4 \equiv 7 \pmod{5}$ (b) $118 \equiv 18 \pmod{5}$ (c) $110 \equiv 93 \pmod{12}$ (d) $63 \equiv 9 \pmod{11}$
- Q 59. Let $a \equiv b \pmod{n}$, $a' \equiv b' \pmod{n}$ and d , in $s N - \{1\}$. Then which of the following need not be true?
 (a) $a + a' \equiv b + b' \pmod{n}$ (b) $aa' \equiv bb' \pmod{n}$
 (c) $a^m \equiv b^m \pmod{n}$ (d) $\frac{a}{d} \equiv \frac{b}{d} \pmod{n}$
- Q 60. If $5x \equiv 2 \pmod{7}$ then
 (a) $x \equiv 6 \pmod{9}$ (b) $x \equiv 6 \pmod{7}$ (c) $x \equiv 7 \pmod{6}$ (d) $x \equiv 2 \pmod{7}$
- The sum of an infinite series $u_1 + u_2 + u_3 + \dots$ will be the finite number l if $\lim_{n \rightarrow \infty} S_n = l$, where $S_n = u_1 + u_2 + u_3 + \dots + u_n$. In such cases it is said that the infinite series is convergent to the sum l . Moreover, a series of positive terms $u_1 + u_2 + u_3 + \dots$ up to ∞ is found to be convergent if
- $$\lim_{n \rightarrow \infty} \frac{u_n}{u_n + 1} > 1.$$
- Q 61. The series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$ to ∞ has the sum
 (a) 1 (b) $\frac{2}{3}$ (c) ∞ (d) 2
- Q 62. The series $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots$ ($x > 0$) is not convergent where x is
 (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) $\frac{2}{3}$
- Q 63. Let $f(n) = \frac{1}{n(n^2 - 1)}$. The series $\sum_{n=1}^{\infty} f(n+1)$ is convergent to the sum

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) 1

(d) $\frac{3}{4}$

• The number of points of intersection of the graphs of the functions $y = f(x)$ and $y = \phi(x)$ give the number of solutions of the equation $f(x) - \phi(x) = 0$. Let $f(x) = ke^x$ and $\phi(x) = x$, where k is a real constant.

Q 64. The line $y = x$ meets the curve $y = ke^x$ for $k \leq 0$ at

(a) no point

(b) one point

(c) two points

(d) more than two points

Q 65. The positive value of k for which $ke^x - x = 0$ has only one real solution is

(a) $\frac{1}{e}$

(b) 1

(c) e

(d) $\log_e 2$

Q 66. For $k > 0$, the set of values of k for which $ke^x - x = 0$ has two distinct real roots is

(a) $\left(0, \frac{1}{e}\right)$

(b) $\left(\frac{1}{e}, 1\right)$

(c) $\left(\frac{1}{e}, \infty\right)$

(d) $(0, 1)$

Answers

1c	2d	3a	4a	5c	6d	7b	8a	9c	10d
11a	12b	13b	14d	15c	16b	17c	18a	19a	20d
21c	22c	23a	24b	25c	26a	27d	28d	29b	30c
31a	32b	33c	34d	35a	36b	37b	38a	39c	40a
41c	42a	43c	44d	45d	46c	47b	48a	49b	50d
51a	52c	53a	54b	55b	56a	57c	58b	59d	60b
61a	62c	63b	64b	65a	66a				

Miscellaneous Questions

Type – 1

Matching Questions

- Q 1. $\lambda x^2 + 2(\lambda - 1)xy + y^2 + 3x + 5y + 1 = 0$ is the equation of
- (i) a circle if λ is (a) 2
- (ii) a parabola if λ is (b) $\frac{1}{2}(3 + \sqrt{5})$
- (iii) an ellipse if λ is (c) 3
- (iv) a hyperbola if λ is (d) 1
- Q 2. Tangents to the parabola $y^2 = 4x$ at the points P and Q meet at R(-1, 0). Then
- (i) the length of the chord PQ is (a) $4\sqrt{2}$
- (ii) the sum of the lengths of the tangents PR and QR is (b) 2
- (iii) the inradius of the ΔPQR is (c) 4
- (iv) the circumradius of the ΔPQR is (d) $2\sqrt{2} - 2$
- Q 3. (i) $\int_{-2}^2 \log(x + \sqrt{1+x^2}) dx$ is equal to (a) 1
- (ii) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left\{ 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right\}$ is (b) 2
- equal to
- (iii) If $\left(\frac{n^2+1}{n+1} - an \right) + 1 = 0$ then a is (c) 0
- (iv) If $y = |\cos x|$ then $\left. \frac{dy}{dx} \right|_{x=3\pi/4}$ is (d) $\frac{1}{\sqrt{2}}$
- Q 4. (i) The maximum value of (a) $-\frac{3}{8}$
- $(2\sin x)(\sqrt{5}\cos x + 2\sin x) - 6$ is
- (ii) $\int_{-2}^2 |x| [x] dx$ is equal to (b) $-\frac{1}{2}$
- (iii) $\lim_{x \rightarrow 0} \frac{\tan^{-1} x - \sin^{-1} x}{x^3}$ is (c) -2
- (iv) If $x^2 y + y^3 = 2$ then $\left. \frac{d^2 y}{dx^2} \right|_{(1,1)}$ is (d) -1
5. (i) If $x^2 - ax + 1 - 2a^2 \geq 0$ for all $x \in \mathbb{R}$ then (a) $-5 < a < 3$
- (ii) If the difference between the roots of $x^2 + ax + 1 = 0$ is less than 5 then (b) $-3 < a < -2$
- (iii) If the complex number $a + 1 + 3i$ for (c) $-\frac{2}{3} \leq a \leq -2$
- all $a \in \mathbb{R}$ is represented by an interior point of the circle $|z| = 5$ then
- (iv) If $\cos^4 x - (a+2)\cos^2 x - (a+3) = 0$ has real solutions for x then (d) $|a| < 3$
6. (i) If $\cos \theta - \sin \theta = \cos \alpha - \sin \alpha$ then the (a) $\frac{\pi}{4}$
- minimum value of $|\theta + \alpha|$ is

- (ii) In a $\triangle ABC$, if $A = 90^\circ$ then (b) $\frac{\pi}{2}$
- $\tan^{-1} \frac{b}{c+a} + \tan^{-1} \frac{c}{a+b}$ is
- (iii) If $x \in [0, \pi]$ and (c) $\frac{\pi}{3}$
- $\log_2(\tan x) + \log_2(\tan 2x) = 0$ then x is
- (iv) In a $\triangle ABC$, if (d) $\frac{\pi}{6}$
- $\frac{\tan A - \tan B}{\tan A + \tan B} = \frac{c-b}{c}$ then A is
- Q 7. (i) $f(x) = x|x-2|$ (a) neither continuous nor differentiable at $x = 2$
 (ii) $g(x) = x^2[x]$ (b) differentiable and continuous for all x
 (iii) $h(x) =$ (c) continuous but not differentiable at $x = 2$
 (iv) $\phi(x) = \begin{cases} [x-1] & \text{for } x \leq 1 \\ x^3 & \text{for } 1 < x \leq 2 \\ 4|x-2| + 2x^2 & \text{for } x > 2 \end{cases}$ (d) differentiable and continuous at $x = 2$
- Q 8. (i) If ${}^nC_{r-1} = 36$, ${}^nC_r = 84$ and ${}^nC_{r+1} = 126$ then r is (a) 6
 (ii) The number of real solutions of the equation $2^x + x^2 = 1$ is (b) 2
 (iii) If the focus of the parabola $x^2 - ky + 3 = 0$ ($k \neq 2$) is $(0, 2)$ then k is (c) 4
 (iv) The number of different ways in which the first twelve natural numbers can be divided in three equal groups such that the numbers in each group are in AP is (d) 3
- Q 9. (i) If z_1, z_2, z_3 , are unimodular complex numbers such that $|z_1 - z_2 + z_3| = 4$ (a) $\frac{1}{2}$
 then $\left| \frac{1}{z_1} - \frac{1}{z_2} + \frac{1}{z_3} \right|$ is (b) 3
 (ii) The planes $2x - 3y - 7z = 0$, $\lambda x - 14y - 13z = 0$ and, $8x - 31y - 33z = 0$ pass through the same line if λ is (c) 2
 (iii) If $\tan^{-1} \frac{\sqrt{1-x^2}-1}{x} : \tan^{-1} x = a : 1$ then a is (d) 4
 (iv) In the $\triangle ABC$ the median $AD = \frac{1}{\sqrt{11-6\sqrt{3}}}$ and the median divides $\angle A$ into angles of 30° and 45° . The length of BC is
- Q 10. (i) The line $\frac{x-1}{l} = \frac{y+1}{4} = \frac{z-2}{2}$ lies on (a) 3
 the plane $2x - y + z - 5 = 0$ if l is (b) 1
 (ii) The latus rectum of the rectangular hyperbola $x - y^2 = 1$ is (c) 4
 (iii) The line cutting a positive intercept 1 on the y -axis will be at antangent to the curve $y = 12x$ if its slope is

- (iv) The lines $\frac{x-5}{4} = \frac{y-7}{m} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ are intersecting lines if m is (d) 2
- Q 11. (i) $\int_0^4 \frac{f(t)}{f(t)+f(4-t)} dt$ is equal to (a) 2
- (ii) If $f(x)$ is an even function then $\int_{-2}^2 \{x^3 f(x) + x f''(x) + 2\} dx$ is equal to (b) 1
- (iii) If $f(x) = \int_0^x |\cos t| dt$ then $f'\left(\frac{\pi}{2}\right)$ equals (c) 0
- (iv) The value of b for which the area bounded by the parabolas $y = x - bx^2$ and $y = \frac{1}{b} x^2$ ($b > 0$) is the maximum is (d) 8
- Q 12. (i) If $f: [2, +\infty) \rightarrow X$ be a bijective, where $f(x) = 5 - 4x + x^2$, then X is (a) $\mathbb{R} - [-2, 2]$
- (ii) The domain of $\phi(x) = \frac{x-3}{(x-1)\sqrt{x^2-4}}$ is (b) $[1, +\infty)$
- (iii) The interval in which $f(x) = 2x^2 - \log x$ is monotonic increasing is (c) $(-2, -1)$
- (iv) $4x^3 + 9x^2 + 6x + 3 = 0$ has a real root lying in (d) $\left(\frac{1}{2}, +\infty\right)$
- Q 13. (i) The focus of the parabola $y^2 - 2y + 8x - 23 = 0$ is (a) $(1, -2)$
- (ii) The centre of the ellipse $9x^2 + 5y^2 - 30y = 0$ is (b) $\left(\frac{3}{2}, -\frac{3}{2}\right)$
- (iii) The vertex of the parabola $x^2 + 2xy + x = -3 + y - y^2$ is (c) $(1, 1)$
- (iv) The tangents at the ends of the double ordinate through the focus of $x^2 = 2(x + 2y) + 3$ meet at (d) $(0, 3)$
- Q 14. (i) If a is a nonreal root of $x^3 - 3x^2 + 3x - 2 = 0$ then $\alpha^{6n} + \alpha^{-6n}$ is equal to (a) 3
- (ii) The order of the differential equation whose general solution is $y = (c_1 x + c_2)e^{ax} + c_3 e^{bx+c_4}$, where c_1, c_2, c_3, c_4 are arbitrary constants and a, b are fixed constants, is (b) 4
- (iii) Let $f(x+y) = f(x) + f(y)$ and $f(1) = 1$. If $\phi(x) = \int_0^x f(2x) dx$ then $\phi'(2)$ is (c) $\frac{4}{3}$
- (iv) If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors where $|\vec{a}| = |\vec{b}| = 2, |\vec{c}| = 1$ then (d) 2

- $\frac{1}{2}[\vec{a} \times \vec{b} \times \vec{b} \times \vec{c} \times \vec{c} \times \vec{a}]$ is
- Q 15. (i) If $|z - 2i| \leq 3$ then the maximum value of $|z + 3|$ is (a) 3
(ii) If the greatest coefficient in the expansion of $(1 + x)^n$ ($n \in \mathbb{N}$) is $\frac{1}{24} n(n-1)(n-2)(n-3)$ then the value of n is (b) 5
(iii) If $\vec{\lambda} = \vec{a} + 2\vec{b} - \vec{c}$, $\vec{\mu} = \vec{a} + \vec{b} + 3\vec{c}$, $\vec{\lambda} = 2\vec{a} + 3\vec{b} + \vec{c}$ and $[\vec{a} \vec{b} \vec{c}] = 3$ then $[\vec{\lambda} \vec{\mu} \vec{v}]$ is (c) 4
(iv) Let $|f(x)| \leq 1$ for $x \in [0, 1)$ and $|f(x)| \leq 2$ for $x \in [1, 3]$ then the greatest value of $\int_0^3 f(x) dx$ is (d) 8
- Q 16. In the parabola $y^2 + 4 = 4x$, a chord passing through the point $(2, 0)$ cuts the parabola at P and Q . If $P = (5, 4)$ and the tangents at P and Q meet at R then match the following.
- (i) The focus is (a) $\left(0, \frac{3}{2}\right)$
(ii) The centroid of the ΔPQR is (b) $(2, 0)$
(iii) The circumcentre of the ΔPQR is (c) $\left(\frac{25}{12}, \frac{3}{2}\right)$
(iv) The orthocentre of the ΔPQR is (d) $\left(\frac{25}{8}, \frac{3}{2}\right)$
- Q 17. (i) $[\vec{a} \vec{b} \vec{i}] \vec{i} + [\vec{a} \vec{b} \vec{j}] \vec{j} + [\vec{a} \vec{b} \vec{k}] \vec{k}$ equals (a) $\vec{a} + \vec{b}$
(ii) If $\vec{a} \times (\vec{a} \times \vec{b}) = \vec{b} \times (\vec{b} \times \vec{c})$ and $\vec{a} \cdot \vec{b} \neq 0$ then $[\vec{a} \vec{b} \vec{c}]$ equals (b) $\vec{a} \times \vec{b}$
(iii) If $|\vec{a}| = |\vec{b}| = 1$, $|\vec{a} \times \vec{b}| = \sqrt{2}$ and $\vec{c} = \vec{a} - \vec{b} + \vec{a} \times \vec{b}$ then $(\vec{a} \times \vec{b}) \times \vec{c}$ equals (c) 0
(iv) If $\vec{a}, \vec{b}$ are orthogonal unit vectors then $[\vec{a} + \vec{a} \times \vec{b} \vec{b} + \vec{a} \times \vec{b} \vec{a} \times \vec{b}]$ equals (d) 1
- Q 18. (i) Given $a^2 + a + 1$ is divisible by 3. When a is divided by 3 the remainder will be (a) 2
(ii) $\lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^2} \cdot \sin(n!)}{n+1}$ is (b) 3
(iii) If $f(n+1) = \frac{1}{2} \left\{ f(n) + \frac{9}{f(n)} \right\}$ for all $n \in \mathbb{N}$ and $f(n) > 0$ for all $n \in \mathbb{N}$ then $\lim_{n \rightarrow \infty} f(n)$ is (c) 0
(iv) If $|\vec{a}| = 1$ then $|\vec{a} \times \vec{i}|^2 + |\vec{a} \times \vec{j}|^2 + |\vec{a} \times \vec{k}|^2$ equals (d) 1
- Q 19. (i) An unbiased dice is thrown and the (a) $\frac{2}{3}$

number shown is put for p in $x^2 + px + 2 = 0$.

The probability of the equation to have real roots is

(ii) $\vec{a}, \vec{b}$ are two unit vectors inclined at $\frac{\pi}{3}$. Then (b) $\frac{2}{5}$

$[\vec{a} \vec{b} + \vec{a} \times \vec{b}]$ is

(iii) If $f(x)$ is differentiable and (c) -1

$\int_0^t xf(x)dx = \frac{2}{5}t^5$ then $f\left(\frac{4}{25}\right)$ equals

Q 20. (i) Let (a) 0

$f(x) = 2\sin^2\beta + 4\cos(x + \beta) \cdot \sin x \cdot \sin \beta + \cos 2(x + \beta)$.

Then the value of $\{f(\alpha)\}^2 + \left\{f\left(\frac{\pi}{4} - \alpha\right)\right\}^2$ is

(ii) n is given a value from the set (b) $\frac{1}{3}$

$\{1, 2, 3, \dots, 12\}$ at random. The probability of the value of $(1 + i)^n$ being real is

(iii) If $[\vec{x} \vec{y} \vec{a}] = [\vec{x} \vec{y} \vec{b}] = [\vec{a} \vec{b} \vec{c}] = 0$, each of the

vectors being a nonzero vector, then $[\vec{x} \vec{y} \vec{c}]$ equals

(iv) If $\frac{2 + \sin x}{y + 1} \frac{dy}{dx} + \cos x = 0$, where $y(0) = 1$, (d) $\frac{1}{4}$

then $y\left(\frac{\pi}{2}\right)$ equals

Q 21. (i) If the vertices of a triangle are represented by the complex numbers 0, z_1 and z_2 such (a) right-angled

that $\frac{z_1}{z_2} + \frac{z_2}{z_1} = 1$ then the triangle is

(ii) If a vertex, the circumcentre and the centroid of a triangle be (0, 0), (3, 4) and respectively, the triangle is (b) equilateral

(iii) If in a $\triangle ABC$, $\sin A \cdot \cos B = 1 - \frac{1}{\sqrt{2}}$ and (c) isosceles

$\tan A \cdot \cot B = \sqrt{2} - 1$ then the triangle is

(iv) The triangle formed by the lines (d) isosceles right-angled

$\frac{x}{3} = \frac{y}{0} = \frac{z}{4}, \frac{x}{0} = \frac{y}{5} = \frac{z}{0}$ and $\frac{x-3}{3} = \frac{y}{-5} = \frac{z-4}{4}$ is

Q 22. (i) If $x^2 - 4x - 2\pi \leq \sin^{-1}(\sin 5)$ then x lies in (a) $\mathbb{R} - [-1, 1]$

(ii) If $f(x) = x^2 - 4$ and $g(x) = f(x) + 5$ such that $f\{g(x)\} > 0$ then x lies in (b) $\mathbb{R} - (-1, 1)$

(iii) If $3\sin \theta + 4\cos \theta = \frac{5(2a-3)}{2-3a}$ has a real (c) $[-4, 4]$

solution for θ then a lies in

(iv) If $4x^4 + 9y^4 = 64$ and $z = \sqrt{3}xy$ then (d) $[-1, 5]$

z lies in

Q 23. (i) The number of sides of the quadrilateral (a) 1

formed by the lines $x^2y^2 + 1 = x^2 + 1 = x^2 + y^2$ that touch the circle $x^2 + y^2 - 2x = 0$ is

(ii) $\vec{b}, \vec{c}$ are orthogonal unit vectors and (b) 0

- $\vec{b} \times \vec{c} = \vec{a}$. Then $[\vec{a} + \vec{b} + \vec{c} \quad \vec{a} + \vec{b} \quad \vec{b} + \vec{c}]$ equals
- (iii) A line having direction ratios 1, -1, 5 is perpendicular to the plane OPQ, where $P = (\lambda, 2, 1)$, $Q = (-2, \mu, 1)$ and O is the origin. Then $\lambda + \mu$ equals
- (iv) If the projection of the vector $\vec{i} + \vec{j} + \vec{k}$ on the vector $a\vec{i} + 4\vec{j} + 5\vec{k}$ be $\frac{1}{3\sqrt{5}}$ then a can have the value
- Q 24. (i) Let $a + b + c + d = 8$ and $abcd = 16$, where a, b, c, d are positive numbers. The value of $a + b$ is
- (ii) The total number of ways of selecting and odd number of things from $(2n + 1)$ different things is 1024. Then n is
- (iii) If $\sqrt{3} \sin x + \cos x - 2 = (y - 1)^2$ for $0 \leq x < 8\pi$ then the number of values of the pair (x, y) is
- (iv) $f(x) = 2x^3 - 3ax^2 + \frac{4}{3}a^2x + 1$ ($a > 0$) attains its maximum and minimum at $x = p$ and $x = p^2$, respectively. Then a equals
- Q 25. If $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$ then the Minimum value of $|z_1 - z_2|$ is
- (ii) If $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(e^x - \cos x)}{x^n}$ is a nonzero finite number then the integer n is
- (iii) If $f(x) = \frac{1-x}{1+x}$ for $x > 0$ then the minimum value of $f\{f(x)\} + f\left\{f\left(\frac{1}{x}\right)\right\}$ is
- (iv) If z is a complex number satisfying $z\bar{z} - 2(2 + \bar{z}) + 3 = 0$ then the greatest value of $|z|$ is
- Q 26. (i) The domain of the real valued function $f(x)$ for which $4^{f(x)} + 4^{1-f(x)} = 4^x$ is
- (ii) A point is moving along the curve $y^3 = 27x$. The interval of values of x in which the ordinate changes faster than the abscissa is
- (iii) The range of values of x for which $x^2 - 1$, $2x + 1$ and $x^2 + x + 1$ will be the lengths of three sides of a triangle is
- (iv) The equation $\sin^{-1}x - 3\sin^{-1}a = 0$ has real solutions for x if a belongs to
- Q 27. (i) The total number of positive integral solutions of $xyz = 18$ is
- (ii) In a ΔABC , the minimum value of $\operatorname{cosec} \frac{A}{2} + \operatorname{cosec} \frac{B}{2} + \operatorname{cosec} \frac{C}{2}$ is
- (iii) If $x^2 + x + 1 = 0$ the value of $\sum_{n=1}^9 \left(x^n + \frac{1}{x^n}\right)^2$

(c) 2

(d) 3

(a) 5

(b) 3

(c) 6

(d) 4

(a) 3

(b) 1

(c) 2

(d) 4

(a) $(1, \infty)$ (b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (c) $[1, \infty]$ (d) $(-1, 1)$

(a) 18

(b) $\frac{3}{2}$

(c) 15

- (iv) The value of $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is (d) 6
- Q 28. If ${}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + {}^{2n+1}C_{n+3} + \dots + {}^{2n+1}C_{2n+1}$ equals 1024 then (a) 16
- (ii) Let $A = \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix}$ and $A^T A = I$. (b) 1
- Then the value of $x^2 + y^2 + z^2$ is
- (iii) In the expansion of $(1+x)^{47}$ the coefficients of $(2r+1)$ th term and the 16th term are equal. Then r is (c) 15
- (iv) Let $\frac{dF(x)}{dx} = \frac{e^{\sin x}}{x}$ for $x > 0$. If $\int_1^4 \frac{e^{\sin x^2}}{x} dx$ equals $\frac{1}{2} \{F(k) - F(1)\}$ then k is (d) 5
- Q 29. (i) If $|\vec{a}| = |\vec{b}| = \sqrt{2}$ Then $[\vec{a} \times \vec{b} \vec{a} \vec{b}] + (\vec{a} \cdot \vec{b})^2$ equals (a) 2
- (ii) The number of points $(a+1, a)$, where $a \in \mathbb{Z}$, lying inside the region bounded by the circles $x^2 + y^2 - 2x - 1 = 0$ and $x^2 + y^2 - 2x - 17 = 0$ is (b) 4
- (iii) The straight line joining the points $(0, 3)$ and $(5, -2)$ is a tangent to the curve $y = \frac{x}{x+1}$. (c) 3
- Then c is
- (iv) A bag contains 8 balls of two colours: red and green. 3 balls are taken out at random. The Probability of getting more red balls than green is $\frac{15}{28}$ and that of getting more green balls than red is $\frac{15}{56}$. The number of green balls is (d) 5
- Q 30. (i) If exactly two real common tangents can be drawn to the circles $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 8x - 8y + 6\lambda = 0$ for $\lambda \in \mathbb{Z}$ then the greatest possible value of λ equals (a) 2
- (ii) $\int_{-2\pi}^{2\pi} \{|\sin x| + |\cos x|\} dx$ equals (b) 3
- (iii) The slope of a curve at (x, y) is $-\frac{x+y}{x}$ and it passes through the points $\left(3, -\frac{1}{6}\right)$ and $(\lambda, 1)$ then one of the values of λ is (c) 1
- (iv) $(\lambda, 6, 2)$ is a point on the plane passing through the line $\frac{x+1}{2} = \frac{y-1}{-1} = \frac{z+3}{1}$ and parallel to the line of intersection of plane $x - y - 5z = 6$ and $3x + 5y + 3z = 4$. Then λ is (d) 16

Answers

1.	(i) d	(ii) b	(iii) c	(iv) a
2.	(i) c	(ii) a	(iii) d	(iv) b
3.	(i) c	(ii) b	(iii) a	(iv) d
4.	(i) d	(ii) c	(iii) b	(iv) a
5.	(i) c	(ii) d	(iii) a	(iv) b
6.	(i) b	(ii) a	(iii) d	(iv) c
7.	(i) c	(ii) a	(iii) b	(iv) d
8.	(i) d	(ii) b	(iii) a	(iv) c
9.	(i) d	(ii) b	(iii) a	(iv) c
10.	(i) b	(ii) d	(iii) a	(iv) c
11.	(i) a	(ii) d	(iii) c	(iv) b
12.	(i) b	(ii) a	(iii) d	(iv) c
13.	(i) c	(ii) d	(iii) b	(iv) a
14.	(i) d	(ii) a	(iii) b	(iv) c
15.	(i) c	(ii) d	(iii) a	(iv) a
16.	(i) b	(ii) c	(iii) d	(iv) a
17.	(i) b	(ii) c	(iii) a	(iv) d
18.	(i) d	(ii) c	(iii) b	(iv) a
19.	(i) a	(ii) d	(iii) b	(iv) c
20.	(i) c	(ii) d	(iii) a	(iv) b
21.	(i) b	(ii) c	(iii) a	(iv) d
22.	(i) d	(ii) a	(iii) b	(iv) c
23.	(i) c	(ii) a	(iii) b	(iv) c
24.	(i) d	(ii) a	(iii) d	(iv) c
25.	(i) c	(ii) a	(iii) c	(iv) a
26.	(i) c	(ii) d	(iii) a	(iv) b
27.	(i) a	(ii) d	(iii) a	(iv) b
28.	(i) d	(ii) b	(iii) a	(iv) a
29.	(i) b	(ii) a	(iii) b	(iv) c
30.	(i) b	(ii) d	(iii) a	(iv) b

Miscellaneous Questions

Type – 1

Fill-in/Completion Questions

- Q 1. If $(1 + x + x^2 + x^3)^7 = \sum_{r=0}^{21} a_r x^r$ then the value of $a_0 + a_4 + a_8 + a_{12} + a_{16} + a_{20}$ is
- Q 2. The number of different ways in which 24 apples can be distributed among 3 boys and 3 girls such that the total number of apples received by boys is double that received by the girls is
- Q 3. If $|\vec{a}| = 3, |\vec{b}| = 4, |\vec{c}| = 5, \vec{a} \perp (\vec{b} + \vec{c}), \vec{b} \perp (\vec{c} + \vec{a})$ and $\vec{c} \perp (\vec{a} + \vec{b})$ then $\sqrt{2} |\vec{a} + \vec{b} + \vec{c}|$ is equal to
- Q 4. Three times the least value of the area bounded by the line $y = mx + 1$ and the parabola $y = x^2 + 2x - 3$, m being a parameter, is
- Q 5. If $f(x)$ and $\phi(x)$ are continuous functions in $[0, 4]$ satisfying $f(x) = f(4 - x)$, $\phi(x) + \phi(4 - x) = 3$ and $\phi(x) + \phi(4 - x) = 3 \int_0^4 f(x) dx$ and $\int_0^4 f(x) \phi(x) dx = 2$ then the value of $\int_0^4 f(x) \phi(x) dx$ is
- Q 6. $\int_{10}^{100} (x - [x]) dx$ is equal to
- Q 7. If $f(x) = \int_a^x \frac{1}{f(x)} dx$ ($x > 0$) and $\int_a^1 \frac{1}{f(x)} dx = \sqrt{2}$ then $f(50)$ is
- Q 8. Consider the closed figure C made by the lines $|x| + |y| = \sqrt{2}$. Let S be the region inside the figure C such that any point in it is nearer to the side $x + y = \sqrt{2}$ than the origin. Then three times the area of S is
- Q 9. If $a_i \geq 0$, where $i = 1, 2, 3, \dots, n$, and $\sum_{i=1}^{n-1} a_i = 20$ then the greatest possible value of $\sum_{i=1}^n a_i a_{i+1}$ is
- Q 10. Let $2^{a_1} + 3^{a_2} + 5^{a_3} + 7^{a_4} + 9^{a_5}$ be divisible by 4, where a_1, a_2, a_3, a_4, a_5 are digits. The largest possible number of values of $(a_1, a_2, a_3, a_4, a_5)$ is
- Q 11. Let $f(x) = ax^4 + bx^3 + cx^2 + dx + \lambda$, where $\frac{f(1)}{1} = \frac{f(2)}{2} = \frac{f(3)}{3}$ and $\lim_{x \rightarrow \infty} \frac{f(x)}{x^4} = 1$. If $f(i) + f(\bar{i}) + k = 0$, where $z = \sqrt{-1}$, then k is
- Q 12. A plane exists which is perpendicular to the line $\frac{x}{3} = \frac{y-1}{5} = \frac{z+1}{7}$ and passes through the line $\frac{x+3}{10} = \frac{y-1}{-20} = \frac{z+1}{k}$. Then k is

1.4096	2.6885	3.10	4.32	5.3
6.45	7.10	8.4.	9.100	10.45000
11.20	12.10			

For example, the answer '4096', i.e., '04096', of **Q.1** can be shown as below by shading:

0	1	2	3	4	5	6	7	8	9
●	○	○	○	○	○	○	○	○	○
○	○	○	○	●	○	○	○	○	○
●	○	○	○	○	○	○	○	○	○
○	○	○	○	○	○	○	○	○	●
○	○	○	○	○	○	●	○	○	○

	0	1	2	3	4	5	6	7	8	9
0										
1										
2										
3										
4										

Miscellaneous Questions

Type – 1

Assertion–reason–Type Questins

• [Each question contains STATEMENT – 1(Assertion) and STATEMENT–2 (Reason). Each question has 4 choice out of which only one is correct.]

- Q 1. The curve C is the locus of a point whose sum of distances from the points S($\sqrt{3}$, 0) and S'($-\sqrt{3}$, 0) is 4.

STATEMENT–1: The curve C cuts off intercept $2\sqrt{3}$ from the line $2y - 1 = 0$.

Because

STATEMENT–2: The equation of the curve C is $\frac{x^2}{4} + 2y^2 = \frac{5}{4}$.

- (a) Statement–1 is True, Statement–2 is True; Statement–2 is Not a explanation for Statement–1
(b) Statement–1 is True, Statement–2 is True; Statement–2 is Not a correct explanation for Statement–2 is False
(c) Statement–1 is True, Statement–2 is False
(d) Statement–1 is False, Statement–2 is true

- Q 2. STATEMENT–1: $\int_0^{\pi/2} \frac{dx}{1 + \tan^3 x} = \frac{\pi}{4}$.

because

STATEMENT–2: $\int_0^a f(x)dx = \int_0^a f(x+a)dx$ where $f(x)$ is an integrable function.

- (a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
(b) Statement–1 is True, Statement–2 is True; Statement–2 is Not a correct explanation for Statement–1
(c) Statement–1 is True, Statement–2 is False
(d) Statement–1 is False, Statement–2 is True

- Q 3. Let $F(x) = \int_1^{x^2} \cos \sqrt{t} dt$.

STATEMENT–1: $F'(x) = \cos x$.

because

STATEMENT–2: If $f(x) = \int_a^x \phi(t)dt$ then $f'(x) = \phi(x)$.

- (a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
(b) Statement–1 is True, Statement–2 is True; Statement–2 is Not a Correct explanation for Statement–1
(c) Statement–1 is True, Statement–2 is False

(d) Statement–1 is False, Statement–2 is True

Q 4. $a_1, a_2, a_3, \dots, a_n$ are in AP.

because

STATEMENT–2: $a_1 + a_n = a_r + a_{n-r+1}$ for $1 \leq r \leq n$.

(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1

(b) Statement–1 is True, Statement–2 is True; Statement–2 is Not a correct explanation for Statement–1

(c) Statement–1 is True, Statement–2 is False

(d) Statement–1 is False, Statement–2 is True

Q 5. STATEMENT–1 $1.3.5. \dots (2n-1) > n^n$, $n \in \mathbb{N}$.

because

STATEMENT–2: The sum of the first n odd natural numbers is equal to n^2 .

(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1

(b) Statement–1 is True, Statement–2 is true; Statement–2 is Not a correct explanation for Statement–1

(c) Statement–1 is True, Statement–2 is False

(d) Statement–1 is True, Statement–2 is False

Q 6. ABCD is a quadrilateral in which a circle is inscribed

STATEMENT–1: The length of the sides of the quadrilateral can be in AP.

because

STATEMENT–2: The lengths of tangents from an external point to a circle are equal.

(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1

(b) Statement–1 is True, Statement–2 is True; Statement–2 is Not a correct explanation for Statements–1

(c) Statement–1 is True, Statement–2 is False

(d) Statement–1 is False, Statement–2 is True

Q 7. a, b, c are three unequal positive numbers.

STATEMENT–1: The product of their sum and the sum of their reciprocals exceeds 9.

because

STATEMENT–2: AM of n positive numbers exceeds their HM.

(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1

(b) Statement–1 is True, Statement–2 is True; Statement–2 is Not a correct explanation for Statement–1

(c) Statement–1 is True, Statement–2 is False

(d) Statement-1 is False, Statement-2 is True

Q 8. z is a unimodular complex number.

STATEMENT-1; $\arg(z^2 + \bar{z}) = \arg z$.

Because

STATEMENT-2: $\bar{z} = \cos(\arg z) - i \sin(\arg z)$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False

(d) Statement-1 is False, Statement-2 is True

Q 9. Let $(1 + x)^{36} = a_0 + a_1x + a_2x^2 + \dots + a_{36}x^{36}$.

STATEMENT-1: $a_0 + a_3 + a_6 + \dots + a_{36} = \frac{2}{3}(2^{35} + 1)$

because

STATEMENT-2: $a_0 + a_1 + a_2 + \dots + a_{36} = 2^{36}$ and
 $a_0 + a_2 + a_4 + \dots + a_{36} = 2^{35}$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False

(d) Statement-1 is False, Statement-2 is True

Q 10. Let a, b, c be real such that $ax^2 + bx + c = 0$ and $x^2 + x + 1 = 0$ have a common root.

STATEMENT-1: $a = b = c$.

because

STATEMENT-2: Two quadratic equations with real coefficients cannot have only one imaginary root common.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False

(d) Statement-1 is False, Statement-2 is True

Q 11. Given in $\triangle ABC$, $a:b:c = \cos A : \cos B : \cos C$.

STATEMENT-1: $\triangle ABC$ is equilateral. because

STATEMENT-2: $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, $\cos B = \frac{c^2 + a^2 - b^2}{2ca}$, $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False

(d) Statement-1 is False, Statement-2 is True

Q 12. STATEMENT-1: $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b} = 0$.

because

STATEMENT-2: If $\vec{p}, \vec{q}, \vec{r}$ are linearly dependent vectors then they are coplanar.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 13. STATEMENT-1: The probability of getting a tail most of the time in 10 tosses of a unbiased coin is

because

STATEMENT-2: ${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_n = 2^{2n-1}$, $n \in \mathbb{N}$.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 14. STATEMENT-1: The probability of solving a new problem by 3 students are α and τ respectively. The probability that the problem will be solved by them is

because

STATEMENT-2: If A, B and C are three independent events then the probability of at least one of them happening = $1 - P(A)P(B)P(C)$.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 15. Let A and B are two independent events. STATEMENT-1: If $P(A) = 0.3$ and $P(A \cup B) = 0.8$ then $P(B)$ is

because

STATEMENT-2: $P(E) = 1 - P(\bar{E})$ where E is any event.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 16. The equations of two straight lines are

$$\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{-3} \text{ and } \frac{x-2}{1} = \frac{y-1}{-3} = \frac{z+3}{2}$$

STATEMENT-1: The given lines are coplanar.

because

STATEMENT-2: The equations

$$2r - s = 1$$

$$r + 3s = 4$$

$$3r + 2s = 5$$

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 17. Given two straight lines whose equations are

STATEMENT-1: The line of shortest distance between the given lines is perpendicular to the plane $x + 3y + 5z = 0$.

because

STATEMENT-2: The direction ratios of the normal to the plane $ax + by + cz + d = 0$ are $\frac{a}{d}, \frac{b}{d}, \frac{c}{d}$.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 18. Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$ where $n \in \mathbb{N}$.

STATEMENT-1: $\int_0^{\pi/4} \tan^4 x \, dx = \frac{3\pi - 8}{12}$.

because

STATEMENT-2: $I_n + I_{n-2} = \frac{1}{n-1}$

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Q 19. STATEMENT-1: The cubic equation $4x^3 - 15x^2 + 14x - 5 = 0$ has a roots in the interval $(2, 3)$

because

STATEMENT-2: If $f(x) = 0$ is a polynomial equation which has two real roots α, β ($\alpha < \beta$) then $f'(x) = 0$ will have a root γ such that $\alpha < \gamma < \beta$.

- (a) Statement-1 is True, Statement-2 is True; Statement is Not a explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Answers

1c	2c	3d	4a	5d	6a	7a	8d	9b	10a
11a	13d	13c	14d	15a	16a	17d	18a	19a	